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=====
Migration 065: Add putt and lie_after columns to golf_shots\n
2-- Purpose: Support
enhanced shot tracking with putt details and lie tracking\n
3--
=====
4--
5--
6--
7--
1. Add putt detail columns to golf_shots\n
8--
=====
9--
10-- Putt distance in feet (converted from yards if needed during entry)\n
11-- ALTER TABLE
golf_shots\n
12-- ADD COLUMN IF NOT EXISTS putt_distance_feet NUMERIC(5,1);\n
13-- COMMENT
ON COLUMN golf_shots.putt_distance_feet IS 'Distance of putt in feet (for putting shots
only)';\n
14--
15-- Whether the putt was made (holed)\n
16-- ALTER TABLE golf_shots\n
17-- ADD COLUMN IF NOT EXISTS putt_made BOOLEAN;\n
18-- COMMENT ON COLUMN golf_shots.putt_made
IS 'Whether the putt was holed (for putting shots only)';\n
19--
20--
=====
21-- 2. Add
lie_after column to track where ball ended up\n
22--
=====
23--
24-- Lie after the shot (where the ball ended up)\n
25-- ALTER TABLE golf_shots\n
26-- ADD
COLUMN IF NOT EXISTS lie_after TEXT;\n
27-- COMMENT ON COLUMN golf_shots.lie_after IS 'Lie
after shot: fairway, rough, sand, green, penalty, recovery';\n
28--
29-- Add check
constraint for valid lie_after values\n
30-- ALTER TABLE golf_shots\n
31-- DROP CONSTRAINT
IF EXISTS golf_shots.lie_after_check;\n
32--
33-- ALTER TABLE golf_shots\n
34-- ADD
CONSTRAINT golf_shots_lie_after_check\n
35-- CHECK (lie_after IS NULL OR lie_after IN ('tee',
'fairway', 'rough', 'sand', 'green', 'penalty', 'recovery', 'other'));\n
36--
37--
=====
38-- 3.
Create indexes for efficient queries\n
39--
=====
40--
41-- Index for putt analysis queries\n
42-- CREATE INDEX IF NOT EXISTS
idx_golf_shots_putt_made\n
43-- ON golf_shots(putt_made) WHERE putt_made IS NOT NULL;\n
44--
45-- Index for shot result analysis by lie\n
46-- CREATE INDEX IF NOT EXISTS
idx_golf_shots_lie_after\n
47-- ON golf_shots(lie_after) WHERE lie_after IS NOT NULL;\n
48--
49-- Composite index for putting stats queries\n
50-- CREATE INDEX IF NOT EXISTS
```

```

idx_golf_shots_putting_analysis\n      51--ON golf_shots(shot_type, putt_made,
putt_distance_feet)\n      52--WHERE shot_type = 'putting';\n      53--\n      54--
===== \n      55-- 4.
Backfill putt_made from existing data where possible\n      56--
===== \n      57--\n
58-- Set putt_made = true for putting shots that resulted in 'hole'\n      59--UPDATE
golf_shots\n      60--SET putt_made = true\n      61--WHERE shot_type = 'putting'\n      62-- AND
result = 'hole'\n      63-- AND putt_made IS NULL;\n      64--\n      65-- Set putt_made = false for
putting shots that didn't result in 'hole'\n      66--UPDATE golf_shots\n      67--SET putt_made =
false\n      68--WHERE shot_type = 'putting'\n      69-- AND result IS NOT NULL\n      70-- AND
result != 'hole'\n      71-- AND putt_made IS NULL;\n      72--\n      73--
===== \n      74-- 5.
Backfill lie_after from result where possible\n      75--
===== \n      76--\n
77-- Map result to lie_after for existing shots\n      78--UPDATE golf_shots\n      79--SET
lie_after = CASE result\n      80--    WHEN 'fairway' THEN 'fairway'\n      81--    WHEN 'rough'
THEN 'rough'\n      82--    WHEN 'sand' THEN 'sand'\n      83--    WHEN 'green' THEN 'green'\n
84--    WHEN 'hole' THEN 'green'\n      85--    WHEN 'penalty' THEN 'penalty'\n      86--    WHEN
'other' THEN 'recovery'\n      87--    ELSE NULL\n      88-- END\n      89--WHERE lie_after IS NULL
AND result IS NOT NULL;\n
90--\n\n\n}}}}, {"uuid": "e2c9dd4a-66c5-45bb-aa50-1803791937c7", "timestamp": "2026-02-05T20:17:37.7
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to golf_shots\n          2--\n          3-- FIX: The application code has been trying to save
distance_to_hole_before and\n          4-- distance_to_hole_after columns to golf_shots, but these
columns were never added\n          5-- to the database schema. This migration adds them.\n
6--\n          7-- These columns are critical for:\n          8-- - Approach efficiency calculations
(strokes to hole out by distance)\n          9-- - Proximity stats (how close approach shots land
to the hole)\n         10-- - Strokes gained calculations\n         11-- - Putting stats (first putt
distance)\n         12--\n         13--
===== \n         14-- 1. Add
distance_to_hole_before column\n         15--
===== \n         16--
Distance from ball to hole BEFORE the shot (in yards or feet, see distance_unit_before)\n
17-- Example: 150 yards for an approach shot, 10 feet for a putt\n         18--ALTER TABLE
golf_shots\n         19--ADD COLUMN IF NOT EXISTS distance_to_hole_before NUMERIC(7,2);\n         20--\n
21--COMMENT ON COLUMN golf_shots.distance_to_hole_before IS\n         22-- 'Distance from ball to

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hole before this shot. Units specified by distance_unit_before (yards or feet);\n      23--\n24-- =====\n2. Add distance_to_hole_after column\n      26--\n===== \n      27--\nDistance from ball to hole AFTER the shot (in yards or feet, see distance_unit_after)\n28-- Example: 15 feet after an approach shot lands on the green\n      29--ALTER TABLE\n      30--ADD COLUMN IF NOT EXISTS distance_to_hole_after NUMERIC(7,2);\n      31--\n      32--COMMENT ON COLUMN golf_shots.distance_to_hole_after IS\n      33-- 'Distance from ball to\n      34-- hole after this shot. Units specified by distance_unit_after (yards or feet);\n      35--\n      36--\n3. Add shot_distance column (if not exists)\n      37--\n===== \n      38-- The\n      39-- actual distance the ball traveled (always in yards)\n      40-- distance_to_hole_before - distance_to_hole_after (accounting for units)\n      41--\n      42--ALTER TABLE\n      43--ADD COLUMN IF NOT EXISTS shot_distance NUMERIC(7,2);\n      44--\n      45--COMMENT ON COLUMN golf_shots.shot_distance IS\n      46-- 'Distance the ball traveled on this\n      47-- shot, in yards';\n      48--\n      49--\n4. Add\n      50-- lie_after column (if not exists)\n      51--\n      52--\n      53-- Derived from result: 'green' result -> 'green' lie, 'fairway' result -> 'fairway' lie, etc.\n      54--\n      55--ALTER TABLE golf_shots\n      56--ADD COLUMN IF NOT EXISTS lie_after TEXT;\n      57--\n      58--COMMENT ON COLUMN golf_shots.lie_after IS\n      59-- 'Where the ball ended up (derived from\n      60-- result): tee, fairway, rough, sand, green, other';\n      61--\n      62--\n      63--\n      64--\n      65--\n      66--\n      67--\n      68--\n      69--\n      70--\n      71--\n      72--\n      73--\n      74--\n      75--\n      76--\n      77--\n      78--\n      79--\n      80--\n      81--\n      82--\n      83--\n      84--\n      85--\n      86--\n      87--\n      88--\n      89--\n      90--\n      91--\n      92--\n      93--\n      94--\n      95--\n      96--\n      97--\n      98--\n      99--\n      100--\n      101--\n      102--\n      103--\n      104--\n      105--\n      106--\n      107--\n      108--\n      109--\n      110--\n      111--\n      112--\n      113--\n      114--\n      115--\n      116--\n      117--\n      118--\n      119--\n      120--\n      121--\n      122--\n      123--\n      124--\n      125--\n      126--\n      127--\n      128--\n      129--\n      130--\n      131--\n      132--\n      133--\n      134--\n      135--\n      136--\n      137--\n      138--\n      139--\n      140--\n      141--\n      142--\n      143--\n      144--\n      145--\n      146--\n      147--\n      148--\n      149--\n      150--\n      151--\n      152--\n      153--\n      154--\n      155--\n      156--\n      157--\n      158--\n      159--\n      160--\n      161--\n      162--\n      163--\n      164--\n      165--\n      166--\n      167--\n      168--\n      169--\n      170--\n      171--\n      172--\n      173--\n      174--\n      175--\n      176--\n      177--\n      178--\n      179--\n      180--\n      181--\n      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```

iku-4-5-20251001","id":"msg_0148M5W9LPa6HZPMxQtZJsBC","type":"message","role":"assistant","content":[{"type":"text","text":"Perfect! Now I have comprehensive information. Let me compile the final report:\n\n## Shot Tracking System Documentation\n\nBased on my thorough investigation of the codebase, here's the complete documentation of the golf shot tracking system:\n\n---\n\n### 1. SHOT TRACKING DATA MODEL\n\n#### Core Tables\n\nStructure:\n\n**golf_shots** (Main shot tracking table)\n\nLocation: `supabase/migrations/021\_golf\_rounds.sql` (initial schema) with enhancements in migrations 040, 064-066\n\nPrimary key: `id` (UUID)\n\nForeign key: `hole\_id` references `golf\_holes(id)` with CASCADE delete\n\nAlso stores: `round\_id`, `hole\_number` as denormalized fields for fallback lookups\n\n**Key Fields in golf_shots:**\n\n- `shot\_number`: INTEGER - Sequential shot number for this hole (1, 2, 3...)\n\n- `shot\_type`: TEXT - 'tee' | 'approach' | 'around_green' | 'putting' | 'penalty'\n\n- `club\_type`: TEXT - 'driver' | 'non_driver' | 'putter' (for strokes gained calculations)\n\n- `lie\_before`: TEXT - Ball position before shot: 'tee' | 'fairway' | 'rough' | 'sand' | 'green' | 'other'\n\n- `distance\_to\_hole\_before`: NUMERIC - Distance to hole before shot (yards or feet)\n\n- `distance\_unit\_before`: TEXT - 'yards' | 'feet'\n\n- `result`: TEXT - Where ball ended up: 'fairway' | 'rough' | 'sand' | 'green' | 'hole' | 'other' | 'penalty'\n\n- `distance\_to\_hole\_after`: NUMERIC - Distance to hole after shot\n\n- `distance\_unit\_after`: TEXT - 'yards' | 'feet'\n\n- `shot\_distance`: NUMERIC - Actual distance ball traveled (calculated)\n\n- `lie\_after`: TEXT - Derived from result (tee|fairway|rough|sand|green|penalty|recovery)\n\n- `putt\_break`: TEXT - 'left_to_right' | 'right_to_left' | 'straight' | 'multiple'\n\n- `putt\_slope`: TEXT - 'uphill' | 'downhill' | 'level' | 'severe'\n\n- `putt\_distance\_feet`: NUMERIC - Distance of putt in feet\n\n- `putt\_made`: BOOLEAN - Whether putt was holed\n\n- `miss\_direction`: TEXT - Legacy field for directional misses\n\n- `is\_penalty`: BOOLEAN - Whether this is a penalty stroke\n\n- `penalty\_type`: TEXT - 'ob' | 'water' | 'unplayable' | 'lost'\n\n- `created\_at`, `updated\_at`: TIMESTAMPTZ\n\n**Related Detail Tables:**\n\n- **putt_details** (Migration 064)\n\n - `shot\_id` (FK to golf_shots)\n\n - `miss\_tags`: TEXT[] - Array of tags: 'low' | 'high' | 'short'\n\n - `break\_direction`: 'left_to_right' | 'right_to_left' | 'straight' | 'multiple'\n\n - `estimated\_break\_inches`: Integer 0-48\n\n - `distance\_feet`: NUMERIC 0-200\n\n - `made`: BOOLEAN\n\n- **approach_miss_details** (Migration 064)\n\n - `shot\_id` (FK to golf_shots)\n\n - `miss\_direction`: 'short' | 'long' | 'left' | 'right' | 'short_left' | 'short_right' | 'long_left' | 'long_right'\n\n - `lie\_type`: 'fairway' | 'rough' | 'sand' | 'recovery' | 'hazard'\n\n - `distance\_from\_green\_yards`: NUMERIC\n\n- **golf_player_stats_cache** (Migration 040 - Performance optimization)\n\n - Caches computed statistics per player to avoid recalculation\n\n - 100+ fields including strokes gained breakdowns, putting percentages, fairway stats, etc.\n\n - Updated after round completion via trigger\n\n- **golf_round_stats_cache** (Migration 040)\n\n - Per-round cached statistics\n\n - Stores complete GolfStats output as `detailed\_stats` JSONB\n\n---\n\n### 2. SHOT TRACKING UI COMPONENTS\n\n**File Structure:** `src/components/golf/`\n\n- **Primary Entry Component:**\n\n - `ShotTrackingComprehensive.tsx` - Main shot-by-shot entry interface\n\n - Manages multiple shots per hole\n\n - Tracks club type, lies, distances, miss directions\n\n - Handles putt classification (miss tags, break direction)\n\n - Handles approach miss classification (direction, lie type)\n\n - Auto-save to Supabase every 30 seconds (configurable)\n\n- **Mobile Entry Interface:**\n\n - `MobileScoreEntry.tsx` - Quick score entry for fast play\n\n - Simplified interface: score, putts, fairway hit, GIR\n\n - Swipe gesture navigation between holes\n\n - Haptic feedback on interactions\n\n - Can switch to detailed shot tracking mode\n\n- **Display Components:**\n\n - `LiveScorecard.tsx` - Shows running scorecard with stats\n\n - Displays holes played, total score, to par, putts, fairways, GIR\n\n - Color-coded hole scores (eagle, birdie, par, bogey, etc.)\n\n - Front 9 / Back 9 grid views\n\n - Tap to navigate between holes\n\n- **Round Management:**\n\n - `SaveRoundModal.tsx` - Options when exiting a round\n\n - "Save for Later" - Partial round saved to database\n\n - "Delete Round" - Discard incomplete round\n\n - Shows current hole progress\n\n - `RoundCompletionSummary.tsx` - Final summary after round\n\n - `ResumeDraftModal.tsx` - Resume previously saved draft round\n\n- **Additional Entry Components:**\n\n - `putt-miss-tag-selector.tsx` - UI for selecting putt miss tags (low/high/short)\n\n - `approach-miss-selector.tsx` - UI for selecting approach miss direction\n\n - `putt-tendencies-card.tsx` - Display putting tendencies analysis\n\n- **Score Review:**\n\n - `RoundScorecard.tsx` - View completed round scorecard\n\n - `ReviewScorecard.tsx` - CoachHelm review interface\n\n---\n\n### 3. HOW SHOTS ARE SAVED - Data Flow\n\n- **UI Flow** → Server Action → Database\n\n- **Client Component:**\n\n - `new-round-client.tsx` - Collects

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hole configuration (course, par, yardage)\n - Uses `ShotTrackingComprehensive` for shot
entry\n - Builds shot data structure: `ShotRecord[]`\n - Calls auto-save via
`useAutoSaveRound` hook (saves incomplete rounds)\n - On completion, calls
`submitGolfRoundComprehensive()`\n\n**2. ShotRecord Structure (from
ShotTrackingComprehensive.tsx):**\n```\ntypescript\ninterface ShotRecord {\n  id?: string;\n  // DB ID for existing shots\n  shotNumber: number;\n  shotType: 'tee' | 'approach' |\n  'around_green' | 'putting' | 'penalty';\n  clubType: 'driver' | 'non_driver' | 'putter';\n  lieBefore: 'tee' | 'fairway' | 'rough' | 'sand' | 'green' | 'other';\n  distanceToHoleBefore: number;\n  distanceUnitBefore: 'yards' | 'feet';\n  result: 'fairway' | 'rough' | 'sand' |\n  'green' | 'hole' | 'other' | 'penalty';\n  distanceToHoleAfter: number;\n  distanceUnitAfter: 'yards' | 'feet';\n  shotDistance: number;\n  missDirection?: string;\n  puttBreak?: 'right_to_left' | 'left_to_right' | 'straight' | 'multiple';\n  puttSlope?: 'uphill' | 'downhill' | 'level' | 'severe';\n  isPenalty: boolean;\n  penaltyType?: 'ob' | 'water' | 'unplayable' | 'lost';\n  puttMissTags?: PuttMissTag[]; // 'low' | 'high' | 'short'\n  puttDistanceFeet?: number;\n  approachMissDirection?: ApproachMissDirection;\n  approachMissLieType?: 'fairway' | 'rough' | 'bunker' | 'hazard';\n}\n```\n\n**3. HoleStats Structure (completed hole):**\n```\ntypescript\ninterface HoleStats {\n  holeNumber: number;\n  par: number;\n  yardage: number;\n  score: number;\n  putts: number;\n  fairwayHit: boolean | null;\n  greenInRegulation: boolean;\n  drivingDistance: number | null;\n  usedDriver: boolean | null;\n  approachDistance: number | null;\n  approachProximity: number | null;\n  approachMissDirection: string | null;\n  scrambleAttempt: boolean;\n  scrambleMade: boolean;\n  sandSaveAttempt: boolean;\n  sandSaveMade: boolean;\n  penaltyStrokes: number;\n  shots: ShotRecord[]; // Array of shots for this hole\n}\n```\n\n**4. Server Action:**\n`submitGolfRoundComprehensive()` (golf.ts, line 569)**\n\nSteps:\n1. Authenticate user via Supabase\n2. Get player record\n3. Verify authorization (player owns round)\n4. Delete existing shots/holes if updating\n5. **Calculate round totals** from holes:\n   - `total_score` = SUM(hole.score)\n   - `total_to_par` = SUM(score - par)\n   - `total_putts` = SUM(hole.putts)\n   - `fairways_hit` = COUNT(fairwayHit=true)\n   - `greens_in_regulation` = COUNT(gir=true)\n6. **Insert golf_rounds record** with:\n   - player_id, course name/location, round_type, round_date\n   - Calculated totals\n   - Status: 'completed'\n7. **Insert golf_holes records** (one per hole):\n   - `sql` INSERT INTO golf_holes (round_id, hole_number, par, score, putts, fairway_hit, gir, penalty_strokes, up_and_down, sand_save)\n8. **Insert golf_shots records** (multiple per hole):\n   - `sql` INSERT INTO golf_shots (round_id, hole_id, hole_number, shot_number, shot_type, club_type, lie_before, distance_to_hole_before, distance_unit_before, result, distance_to_hole_after, distance_unit_after, shot_distance, lie_after, putt_break, putt_slope, putt_distance_feet, putt_made, is_penalty, penalty_type)\n9. **Insert putt_details** for putting shots with miss tags\n10. **Insert approach_miss_details** for approach shots with miss direction\n11. **Invalidate stats cache** via `invalidateOnRoundComplete()`\n12. **Revalidate paths** to refresh UI\n\n**5. Partial Round Save (Draft):**\n- `savePartialRound()` - Saves incomplete round data\n- Uses `golf_round_drafts` table for drafts\n- Auto-triggered every 30 seconds during play\n- Can resume from draft later\n\n**6. Shot Update/Delete (Individual shot edits):**\n- `updateShot()` - Modify existing shot\n- `deleteShot()` - Remove shot (triggers `resequence_golf_shots_after_delete()`) - Auto-renumbers remaining shots to maintain contiguous sequence\n\n---\n\n### 4. SHOT DATA SCHEMA - Complete Field Reference\n\n**What Gets Stored Per Shot:**\n\n| Field | Type | Purpose | Example |\n|-----|-----|-----|-----|\n| `shot_number` | INT | Sequential in hole | 1, 2, 3, 4... |\n| `shot_type` | TEXT | Classification | 'tee', 'approach', 'putting' |\n| `club_type` | TEXT | Club category | 'driver', 'non_driver', 'putter' |\n| `lie_before` | TEXT | Ball position before | 'tee', 'fairway', 'rough' |\n| `distance_to_hole_before` | NUMERIC | Distance before shot | 150 (yards), 10 (feet) |\n| `distance_unit_before` | TEXT | Unit of measurement | 'yards', 'feet' |\n| `result` | TEXT | Where ball landed | 'fairway', 'green', 'rough' |\n| `distance_to_hole_after` | NUMERIC | Distance after shot | 15 (remaining distance) |\n| `distance_unit_after` | TEXT | Unit after | 'yards', 'feet' |\n| `shot_distance` | NUMERIC | Ball travel distance | 135 (yards) |\n| `lie_after` | TEXT | Derived from result | 'fairway', 'green' |\n| `putt_break` | TEXT | Break direction (putts only) | 'left_to_right', 'straight' |\n| `putt_slope` | TEXT | Slope category (putts) |

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`putt`, 'downhill' |\n| `putt_distance_feet` | NUMERIC | Putt distance (feet) | 8.5 |\n|
`putt_made` | BOOLEAN | Putt holed | true/false |\n| `miss_direction` | TEXT | Legacy miss
field | 'short', 'left' |\n| `is_penalty` | BOOLEAN | Penalty stroke flag | false |\n|
`penalty_type` | TEXT | Type of penalty | 'ob', 'water' |\n| `puttMissTags` | TEXT[] | Putt
miss classification | ['low', 'short'] |\n| `approachMissDirection` | TEXT | Approach miss
direction | 'short_left', 'long_right' |\n| `approachMissLieType` | TEXT | Lie type after
approach miss | 'rough', 'bunker' |\n\n---\n\n### 5. ROUND SUBMISSION FLOW\n\n**Complete
Lifecycle:**\n\n```\nUser starts round\n ↓\nEnters hole configuration (par, yardage per
hole)\n ↓\nFor each hole:\n - Enters score, putts, fairway hit, GIR via MobileScoreEntry
OR\n - Tracks each shot individually via ShotTrackingComprehensive\n - (Auto-save happens
every 30 sec to golf_round_drafts)\n ↓\nAfter last hole is completed:\n - Shows
RoundCompletionSummary\n - User confirms submission\n ↓\nsubmitGolfRoundComprehensive()
called:\n 1. Insert golf_rounds record\n 2. Insert golf_holes (1 per hole, 18 total)\n 3.
Insert golf_shots (1-N per hole)\n 4. Insert putt_details (optional, for putt
classifications)\n 5. Insert approach_miss_details (optional, for approach misses)\n
↓\nStats cache invalidated → triggers recalculation\n ↓\nRound marked as 'completed'\n
↓\nAvailable in dashboards and analytics\n```\n\n**Draft Round Persistence:**\n-
`checkForDraft()` - On app load, check for saved draft\n- `ResumeDraftModal` - Offer to resume
or delete\n- `clearRoundDraft()` - Clean up after submission\n\n---\n\n### 6. DETAILED SCHEMA
RELATIONSHIPS\n\n```\ngolf_rounds (1)\n └─ golf_holes (18) - One per hole\n      └─
golf_shots (1-N) - Multiple shots per hole\n          └─ putt_details (0-1) - Optional, for
putting shots\n              └─ approach_miss_details (0-1) - Optional, for approach
shots\n\n```\n\n**Keys:**\n- `golf_holes.round_id` → `golf_rounds.id`\n- `golf_shots.hole_id` →
`golf_holes.id`\n- `golf_shots.round_id` → `golf_rounds.id` (denormalized for fallback)\n-
`putt_details.shot_id` → `golf_shots.id`\n- `approach_miss_details.shot_id` →
`golf_shots.id`\n\n**Cascade Behavior:**\n- Delete hole → cascades delete all shots for that
hole\n- Delete shot → trigger auto-renumbers remaining shots\n- Delete round → cascades delete
all holes and shots\n\n---\n\n### 7. KEY FILES REFERENCE\n\n| File | Purpose
|\n|-----|-----|\n| `src/lib/types/golf.ts` | All TypeScript type definitions (GolfShot,
ShotRecord, etc.) |\n| `src/components/golf/ShotTrackingComprehensive.tsx` | Main detailed
shot entry interface |\n| `src/components/golf/MobileScoreEntry.tsx` | Quick score entry mode
|\n| `src/components/golf/LiveScorecard.tsx` | Running scorecard display |\n|
`src/app/golf/actions/golf.ts` | Server actions: submitGolfRoundComprehensive, updateShot,
deleteShot |\n| `src/app/golf/(dashboard)/dashboard/rounds/new/new-round-client.tsx` | Round
start flow |\n| `supabase/migrations/021_golf_rounds.sql` | Core golf_rounds, golf_holes,
golf_shots tables |\n| `supabase/migrations/040_golf_shot_system.sql` | Stats cache tables,
shot enhancements |\n| `supabase/migrations/047_shot_resequencing.sql` | Auto-resequencing
function after shot deletion |\n| `supabase/migrations/064_putt_approach_details.sql` |
putt_details and approach_miss_details tables |\n|
`supabase/migrations/065_golf_shots_putt_lie_columns.sql` | Added putt_distance_feet,
putt_made, lie_after |\n| `supabase/migrations/066_golf_shots_distance_columns.sql` | Added
distance_to_hole_before/after columns |\n\nThis shot tracking system is comprehensive,
supporting detailed stroke-by-stroke analysis with support for putt classification, approach
miss patterns, penalty tracking, and automatic statistics caching for performance
optimization."}], "stop_reason": null, "stop_sequence": null, "usage": {"input_tokens": 2, "cache_crea
tion_input_tokens": 5391, "cache_read_input_tokens": 144277, "cache_creation": {"ephemeral_5m_input
_tokens": 0, "ephemeral_1h_input_tokens": 5391}, "output_tokens": 1, "service_tier": "standard", "infe
rence_geo": "not_available"}}, "requestId": "req_011CXqYDeDMuiEiZucSGmKpD", "type": "assistant", "uu
id": "eea4c0b8-d343-4b6d-b375-ddc5ea1e885f", "timestamp": "2026-02-05T20:18:15.976Z"}</output>

```

TaskOutput

```
<retrieval status>success</retrieval status><task id>a218c55</task id>
```

Show less

```
<task_type>local_agent</task_type><status>completed</status><output>[Truncated. Full output:
/private/tmp/claude-501/-Users-ricknini-Downloads-helmv3/tasks/a218c55.output]d<LieType,
Recordnumber, number>;\n      34→\n      35→/**\n      36→ * Internal shot representation for
strokes gained calculations\n      37→ * Maps from database fields to calculation-friendly
names\n      38→ */\n      39→export interface SGShot {\n      40→ hole_id: string | null;\n      41→
shot number: number;\n      42→ starting lie: LieType;\n      43→ starting distance yards:
```



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number;\n    44→ ending_lie: LieType;\n    45→ ending_distance_yards: number;\n    46→
is_holed: boolean;\n    47→ is_penalty: boolean;\n    48→}\n    49→\n    50→/**\n    51→ *
Internal hole representation for strokes gained calculations\n    52→ * Maps from database
fields to calculation-friendly names\n    53→ */\n    54→export interface SGHole {\n    55→
id: string;\n    56→ par: number;\n    57→ score: number;\n    58→ putts: number;\n    59→
fairway_hit: boolean | null;\n    60→ green_in_regulation: boolean | null;\n    61→
up_and_down: boolean | null;\n    62→ sand_save: boolean | null;\n    63→ shots?:
SGShot[];\n    64→}\n    65→\n    66→/**\n    67→ * Internal round representation for strokes
gained calculations\n    68→ */\n    69→export interface SGRound {\n    70→ holes:
SGHole[];\n    71→}\n    72→\n    73→/**\n    74→ * Convert database shot to internal shot
format\n    75→ */\n    76→export function convertDbShot(shot: DbGolfShot): SGShot {\n    77→
return {\n    78→   hole_id: shot.hole_id,\n    79→   shot_number: shot.shot_number,\n    80→
   starting_lie: mapLieType(shot.lie_before),\n    81→   starting_distance_yards:
shot.distance_to_hole_before ?? 0,\n    82→   ending_lie: mapLieType(shot.lie_after),\n    83→
   ending_distance_yards: shot.distance_to_hole_after ?? 0,\n    84→   is_holed:
shot.putt_made === true || shot.result === 'holed' || shot.result === 'hole',\n    85→
   is_penalty: shot.is_penalty === true,\n    86→ };}\n    87→\n    88→\n    89→/**\n    90→ *
Convert database hole to internal hole format\n    91→ */\n    92→export function
convertDbHole(hole: DbGolfHole, shots?: DbGolfShot[]): SGHole {\n    93→ return {\n    94→
id: hole.id,\n    95→   par: hole.par,\n    96→   score: hole.score ?? 0,\n    97→   putts:
hole.putts ?? 0,\n    98→   fairway_hit: hole.fairway_hit,\n    99→   green_in_regulation:
hole.gir,\n    100→   up_and_down: hole.up_and_down,\n    101→   sand_save: hole.sand_save,\n    102→
   shots: shots?.map(convertDbShot),\n    103→ };}\n    104→}\n    105→\n    106→/**\n    107→
* Map database lie string to LieType enum\n    108→ */\n    109→export function mapLieType(lie:
string | null): LieType {\n    110→ if (!lie) return 'fairway';\n    111→ const normalized =
lie.toLowerCase();\n    112→ if (normalized === 'tee' || normalized === 'teebox') return
'tee';\n    113→ if (normalized === 'fairway') return 'fairway';\n    114→ if (normalized ===
'rough' || normalized === 'primary_rough' || normalized === 'deep_rough') return 'rough';\n    115→
if (normalized === 'sand' || normalized === 'bunker' || normalized ===
'greenside_bunker' || normalized === 'fairway_bunker') return 'sand';\n    116→ if (normalized
=== 'green' || normalized === 'fringe') return 'green';\n    117→ if (normalized ===
'recovery' || normalized === 'hazard' || normalized === 'penalty') return 'recovery';\n    118→
return 'fairway'; // Default fallback\n    119→}\n    120→\n    121→/**\n    122→ * Strokes gained
result with breakdown by category\n    123→ */\n    124→export interface StrokesGainedResult {\n    125→
sg_off_tee: number;\n    126→ sg_approach: number;\n    127→ sg_around_green: number;\n    128→
sg_putting: number;\n    129→ sg_total: number;\n    130→}\n    131→\n    132→/**\n    133→
* Extended strokes gained breakdown with shot counts and per-shot averages\n    134→ */\n    135→export interface StrokesGainedBreakdown extends StrokesGainedResult {\n    136→
shots_off_tee: number;\n    137→ shots_approach: number;\n    138→ shots_around_green:
number;\n    139→ shots_putting: number;\n    140→ sg_per_shot_off_tee: number;\n    141→
sg_per_shot_approach: number;\n    142→ sg_per_shot_around_green: number;\n    143→
sg_per_shot_putting: number;\n    144→}\n    145→\n    146→// Re-export database types for
external use\n    147→export type { DbGolfShot as GolfShot, DbGolfHole as GolfHole, DbGolfRound
as GolfRound };\n    148→\n    149→// =====\n    150→//
PGA TOUR BASELINE DATA\n    151→// =====\n    152→\n    153→/**\n    154→ * Expected strokes
to hole out from various lies and distances\n    155→ *
Based on PGA Tour Shotlink data\n    156→ * Distance for tee/fairway/rough is in YARDS\n    157→
* Distance for green is in FEET\n    158→ */\n    159→export const PGA_BASELINE_DATA:
BaselineData = {\n    160→   tee: {\n    161→     // Par 5 tee shots (long)\n    162→     600: 4.90,
580: 4.85, 560: 4.80, 540: 4.75, 520: 4.70,\n    163→     500: 4.65, 480: 4.60, 460: 4.55, 440:
4.50, 420: 4.45,\n    164→     // Par 4 tee shots\n    165→     400: 4.10, 380: 4.02, 360: 3.95,
340: 3.88, 320: 3.82,\n    166→     300: 3.75, 280: 3.68, 260: 3.62,\n    167→   },\n    168→
   fairway: {\n    169→     // Long approach\n    170→     275: 3.45, 250: 3.30, 225: 3.15, 200:
2.99, 190: 2.92,\n    171→     180: 2.86, 175: 2.82, 170: 2.78, 165: 2.75, 160: 2.71,\n    172→
     // Mid approach\n    173→     155: 2.68, 150: 2.67, 145: 2.64, 140: 2.61, 135: 2.58,\n    174→
     130: 2.55, 125: 2.53, 120: 2.50, 115: 2.47, 110: 2.44,\n    175→     // Short approach\n    176→
     105: 2.42, 100: 2.40, 95: 2.37, 90: 2.35, 85: 2.32,\n    177→     80: 2.30, 75: 2.27, 70: 2.25,
65: 2.22, 60: 2.20,\n    178→     55: 2.17, 50: 2.15, 45: 2.12, 40: 2.10, 35: 2.08,\n    179→

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30: 2.06, 25: 2.04, 20: 2.02,\n 180→ },\n 181→ rough: {\n 182→ // Rough is harder
than fairway - add penalty\n 183→ 275: 3.65, 250: 3.48, 225: 3.32, 200: 3.15, 190:
3.08,\n 184→ 180: 3.01, 175: 2.98, 170: 2.94, 165: 2.91, 160: 2.87,\n 185→ 155:
2.84, 150: 2.82, 145: 2.79, 140: 2.76, 135: 2.73,\n 186→ 130: 2.70, 125: 2.68, 120: 2.65,
115: 2.62, 110: 2.59,\n 187→ 105: 2.56, 100: 2.54, 95: 2.51, 90: 2.48, 85: 2.45,\n 188→
80: 2.42, 75: 2.40, 70: 2.37, 65: 2.34, 60: 2.32,\n 189→ 55: 2.29, 50: 2.26, 45: 2.24,
40: 2.21, 35: 2.18,\n 190→ 30: 2.16, 25: 2.14, 20: 2.12,\n 191→ },\n 192→ sand: {\n
193→ // Greenside bunker shots (distance in yards)\n 194→ 40: 2.65, 35: 2.58, 30:
2.53, 28: 2.48, 26: 2.44,\n 195→ 24: 2.40, 22: 2.36, 20: 2.32, 18: 2.28, 16: 2.24,\n
196→ 14: 2.20, 12: 2.18, 10: 2.18, 8: 2.14, 6: 2.10,\n 197→ 5: 2.08, 4: 2.06, 3:
2.04,\n 198→ },\n 199→ green: {\n 200→ // Putting distances in FEET\n 201→ 90:
2.40, 80: 2.30, 70: 2.20, 60: 2.15, 55: 2.08,\n 202→ 50: 2.02, 45: 2.00, 40: 1.93, 35:
1.87, 30: 1.82,\n 203→ 28: 1.78, 26: 1.75, 24: 1.72, 22: 1.68, 20: 1.63,\n 204→ 18:
1.58, 16: 1.53, 14: 1.47, 12: 1.42, 10: 1.32,\n 205→ 9: 1.28, 8: 1.24, 7: 1.20, 6: 1.16,
5: 1.12,\n 206→ 4: 1.08, 3: 1.04, 2: 1.00, 1: 1.00, 0: 0,\n 207→ },\n 208→
recovery: {\n 209→ // Trees, deep rough, etc.\n 210→ 200: 3.80, 175: 3.55, 150:
3.35, 125: 3.15, 100: 2.95,\n 211→ 75: 2.75, 50: 2.55, 40: 2.45, 30: 2.35, 20: 2.25,\n
212→ },\n 213→};\n 214→\n 215→// =====\n
216→// HELPER FUNCTIONS\n 217→// =====\n 218→\n
219→/**\n 220→ * Get expected strokes from a given lie and distance\n 221→ * Uses linear
interpolation between known data points\n 222→ */\n 223→export function
getExpectedStrokes(\n 224→ lie: LieType,\n 225→ distanceYards: number,\n 226→
isOnGreen: boolean = false\n 227→): number {\n 228→ const baseline =
PGA_BASELINE_DATA[lie];\n 229→ if (!baseline) return 3.0; // Default fallback\n 230→\n
231→ // For putting, convert yards to feet\n 232→ const distance = isOnGreen || lie ===
'green'\n 233→ ? Math.round(distanceYards * 3) // Convert yards to feet\n 234→ :
distanceYards;\n 235→\n 236→ // Get sorted distances from baseline\n 237→ const
distances = Object.keys(baseline)\n 238→ .map(Number)\n 239→ .sort((a, b) => b - a);
// Sort descending\n 240→\n 241→ // Handle empty baseline\n 242→ if (distances.length
=== 0) {\n 243→ return 3.0;\n 244→ }\n 245→\n 246→ const maxDist =
distances[0]!;\n 247→ const minDist = distances[distances.length - 1]!;\n 248→\n 249→
// If distance is beyond max, extrapolate\n 250→ if (distance >= maxDist) {\n 251→
return baseline[maxDist] ?? 3.0;\n 252→ }\n 253→\n 254→ // If distance is below min,
use minimum\n 255→ if (distance <= minDist) {\n 256→ return baseline[minDist] ??
3.0;\n 257→ }\n 258→\n 259→ // Find surrounding data points and interpolate\n 260→
for (let i = 0; i < distances.length - 1; i++) {\n 261→ const upperDist =
distances[i]!;\n 262→ const lowerDist = distances[i + 1]!;\n 263→ if (distance <=
upperDist && distance >= lowerDist) {\n 264→ const upperStrokes = baseline[upperDist]
?? 3.0;\n 265→ const lowerStrokes = baseline[lowerDist] ?? 3.0;\n 266→\n 267→
// Linear interpolation\n 268→ const ratio = (distance - lowerDist) / (upperDist -
lowerDist);\n 269→ return lowerStrokes + ratio * (upperStrokes - lowerStrokes);\n
270→ }\n 271→ }\n 272→\n 273→ // Fallback - shouldn't reach here\n 274→ return
baseline[minDist] ?? 3.0;\n 275→}\n 276→\n 277→/**\n 278→ * Calculate strokes gained
for a single shot\n 279→ */\n 280→export function calculateShotStrokesGained(shot:
SGShot): number {\n 281→ const expectedBefore = getExpectedStrokes(\n 282→
shot.starting_lie,\n 283→ shot.starting_distance_yards,\n 284→ shot.starting_lie ===
'green'\n 285→ );\n 286→\n 287→ let expectedAfter: number;\n 288→ if
(shot.is_holed) {\n 289→ expectedAfter = 0;\n 290→ } else {\n 291→ expectedAfter
= getExpectedStrokes(\n 292→ shot.ending_lie,\n 293→
shot.ending_distance_yards,\n 294→ shot.ending_lie === 'green'\n 295→ );\n 296→
}\n 297→\n 298→ // SG = Expected(before) - Expected(after) - 1\n 299→ // Positive =
better than average, Negative = worse than average\n 300→ return expectedBefore -
expectedAfter - 1;\n 301→}\n 302→\n 303→/**\n 304→ * Categorize a shot into SG
category\n 305→ */\n 306→export function categorizeShot(shot: SGShot, holePar: number):
keyof StrokesGainedResult | null {\n 307→ // Penalty shots don't contribute to SG
categories meaningfully\n 308→ if (shot.is_penalty) return null;\n 309→\n 310→ //
Putting\n 311→ if (shot.starting_lie === 'green') {\n 312→ return 'sg_putting';\n
313→ }\n 314→\n 315→ // Off the Tee (first shot on par 4 or par 5)\n 316→ if

```



```
shot.number == 1 & holePar >= 4) {\n      317→ return 'sg_off_tee';\n    }\n  319→\n    // Around the Green (within 30 yards, not on green)\n    321→ // Note: green case already handled above\n    322→ if (shot.starting_distance_yards <= 30) {\n    323→ return 'sg_around_green';\n    324→ }\n    325→\n    326→ // Approach (everything else)\n    327→ return 'sg_approach';\n    328→}\n    329→\n    330→// =====\n    331→// MAIN CALCULATION FUNCTIONS\n    332→// =====\n    333→\n    334→/**\n    335→ * Calculate strokes gained for an array of shots\n    336→ */\n    337→export function calculateStrokesGained(\n    338→ shots: SGShot[],\n    339→ holes: SGHole[]\n    340→): StrokesGainedResult {\n    341→ const result: StrokesGainedResult = {\n    342→   sg_off_tee: 0,\n    343→   sg_approach: 0,\n    344→   sg_around_green: 0,\n    345→   sg_putting: 0,\n    346→   sg_total: 0,\n    347→ }; \n    348→\n    349→ // Create hole lookup for par values\n    350→ const holeMap = new Map(holes.map(h => [h.id, h]));\n    351→\n    352→ for (const shot of shots) {\n    353→ const hole = shot.hole_id ? holeMap.get(shot.hole_id) : null;\n    354→ if (!hole) continue;\n    355→\n    356→ const category = categorizeShot(shot, hole.par);\n    357→ if (!category) continue;\n    358→\n    359→ const sg = calculateShotStrokesGained(shot);\n    360→ result[category] += sg;\n    361→ }\n    362→\n    363→ result.sg_total =\n    364→ result.sg_off_tee +\n    365→ result.sg_approach +\n    366→ result.sg_around_green +\n    367→ result.sg_putting;\n    368→\n    369→ return result;\n    370→}\n    371→\n    372→/**\n    373→ * Calculate detailed strokes gained breakdown with per-shot averages\n    374→ */\n    375→export function calculateStrokesGainedBreakdown(\n    376→ shots: SGShot[],\n    377→ holes: SGHole[]\n    378→): StrokesGainedBreakdown {\n    379→ const counts = {\n    380→ off_tee: 0,\n    381→ approach: 0,\n    382→ around_green: 0,\n    383→ putting: 0,\n    384→ };\n    385→\n    386→ const result: StrokesGainedResult = {\n    387→   sg_off_tee: 0,\n    388→   sg_approach: 0,\n    389→   sg_around_green: 0,\n    390→   sg_putting: 0,\n    391→   sg_total: 0,\n    392→ }; \n    393→\n    394→ const holeMap = new Map(holes.map(h => [h.id, h]));\n    395→\n    396→ for (const shot of shots) {\n    397→ const hole = shot.hole_id ? holeMap.get(shot.hole_id) : null;\n    398→ if (!hole) continue;\n    399→\n    400→ const category = categorizeShot(shot, hole.par);\n    401→\n    402→ result[category] += {\n    403→   count: 1,\n    404→   avg: shot.starting_distance_yards / hole.par,\n    405→   diff: shot.starting_distance_yards - hole.par,\n    406→   score: shot.score,\n    407→   par: hole.par,\n    408→   holeId: shot.hole_id,\n    409→   holeName: hole.name,\n    410→   holeType: hole.type,\n    411→   holeOrder: hole.order,\n    412→   holeColor: hole.color,\n    413→   holeImage: hole.image,\n    414→   holeIcon: hole.icon,\n    415→   holeStatus: hole.status,\n    416→   holeNotes: hole.notes,\n    417→   holeDescription: hole.description,\n    418→   holeInstructions: hole.instructions,\n    419→   holeWarnings: hole.warnings,\n    420→   holeTips: hole.tips,\n    421→   holeTricks: hole.tricks,\n    422→   holeSecrets: hole.secrets,\n    423→   holeMysteries: hole.mysteries,\n    424→   holeLegends: hole.legend,\n    425→   holeFolklore: hole.folklore,\n    426→   holeMyths: hole.myths,\n    427→   holeStories: hole.stories,\n    428→   holeAnecdotes: hole.anecdotes,\n    429→   holeCuriosities: hole.curiosities,\n    430→   holeTrivia: hole.trivia,\n    431→   holeFunFacts: hole.fun_facts,\n    432→   holeInterestingFacts: hole.interesting_facts,\n    433→   holeAmazingFacts: hole.amazing_facts,\n    434→   holeIncredibleFacts: hole.incredible_facts,\n    435→   holeUnbelievableFacts: hole.unbelievable_facts,\n    436→   holeMindBlowingFacts: hole.mind_blowing_facts,\n    437→   holeBreathtakingFacts: hole.breathtaking_facts,\n    438→   holeStunningFacts: hole.stunning_facts,\n    439→   holeAmazingStories: hole.amazing_stories,\n    440→   holeIncredibleStories: hole.incredible_stories,\n    441→   holeUnbelievableStories: hole.unbelievable_stories,\n    442→   holeMindBlowingStories: hole.mind_blowing_stories,\n    443→   holeBreathtakingStories: hole.breathtaking_stories,\n    444→   holeStunningStories: hole.stunning_stories,\n    445→   holeAmazingAnecdotes: hole.amazing_anecdotes,\n    446→   holeIncredibleAnecdotes: hole.incredible_anecdotes,\n    447→   holeUnbelievableAnecdotes: hole.unbelievable_anecdotes,\n    448→   holeMindBlowingAnecdotes: hole.mind_blowing_anecdotes,\n    449→   holeBreathtakingAnecdotes: hole.breathtaking_anecdotes,\n    450→   holeStunningAnecdotes: hole.stunning_anecdotes,\n    451→   holeAmazingCuriosities: hole.amazing_curiosities,\n    452→   holeIncredibleCuriosities: hole.incredible_curiosities,\n    453→   holeUnbelievableCuriosities: hole.unbelievable_curiosities,\n    454→   holeMindBlowingCuriosities: hole.mind_blowing_curiosities,\n    455→   holeBreathtakingCuriosities: hole.breathtaking_curiosities,\n    456→   holeStunningCuriosities: hole.stunning_curiosities,\n    457→   holeAmazingTrivia: hole.amazing_trivia,\n    458→   holeIncredibleTrivia: hole.incredible_trivia,\n    459→   holeUnbelievableTrivia: hole.unbelievable_trivia,\n    460→   holeMindBlowingTrivia: hole.mind_blowing_trivia,\n    461→   holeBreathtakingTrivia: hole.breathtaking_trivia,\n    462→   holeStunningTrivia: hole.stunning_trivia,\n    463→   holeAmazingFunFacts: hole.amazing_fun_facts,\n    464→   holeIncredibleFunFacts: hole.incredible_fun_facts,\n    465→   holeUnbelievableFunFacts: hole.unbelievable_fun_facts,\n    466→   holeMindBlowingFunFacts: hole.mind_blowing_fun_facts,\n    467→   holeBreathtakingFunFacts: hole.breathtaking_fun_facts,\n    468→   holeStunningFunFacts: hole.stunning_fun_facts,\n    469→   holeAmazingInterestingFacts: hole.amazing_interesting_facts,\n    470→   holeIncredibleInterestingFacts: hole.incredible_interesting_facts,\n    471→   holeUnbelievableInterestingFacts: hole.unbelievable_interesting_facts,\n    472→   holeMindBlowingInterestingFacts: hole.mind_blowing_interesting_facts,\n    473→   holeBreathtakingInterestingFacts: hole.breathtaking_interesting_facts,\n    474→   holeStunningInterestingFacts: hole.stunning_interesting_facts,\n    475→   holeAmazingAmazingFacts: hole.amazing_amazing_facts,\n    476→   holeIncredibleAmazingFacts: hole.incredible_amazing_facts,\n    477→   holeUnbelievableAmazingFacts: hole.unbelievable_amazing_facts,\n    478→   holeMindBlowingAmazingFacts: hole.mind_blowing_amazing_facts,\n    479→   holeBreathtakingAmazingFacts: hole.breathtaking_amazing_facts,\n    480→   holeStunningAmazingFacts: hole.stunning_amazing_facts,\n    481→   holeAmazingAmazingStories: hole.amazing_amazing_stories,\n    482→   holeIncredibleAmazingStories: hole.incredible_amazing_stories,\n    483→   holeUnbelievableAmazingStories: hole.unbelievable_amazing_stories,\n    484→   holeMindBlowingAmazingStories: hole.mind_blowing_amazing_stories,\n    485→   holeBreathtakingAmazingStories: hole.breathtaking_amazing_stories,\n    486→   holeStunningAmazingStories: hole.stunning_amazing_stories,\n    487→   holeAmazingAmazingAnecdotes: hole.amazing_amazing_anecdotes,\n    488→   holeIncredibleAmazingAnecdotes: hole.incredible_amazing_anecdotes,\n    489→   holeUnbelievableAmazingAnecdotes: hole.unbelievable_amazing_anecdotes,\n    490→   holeMindBlowingAmazingAnecdotes: hole.mind_blowing_amazing_anecdotes,\n    491→   holeBreathtakingAmazingAnecdotes: hole.breathtaking_amazing_anecdotes,\n    492→   holeStunningAmazingAnecdotes: hole.stunning_amazing_anecdotes,\n    493→   holeAmazingAmazingCuriosities: hole.amazing_amazing_curiosities,\n    494→   holeIncredibleAmazingCuriosities: hole.incredible_amazing_curiosities,\n    495→   holeUnbelievableAmazingCuriosities: hole.unbelievable_amazing_curiosities,\n    496→   holeMindBlowingAmazingCuriosities: hole.mind_blowing_amazing_curiosities,\n    497→   holeBreathtakingAmazingCuriosities: hole.breathtaking_amazing_curiosities,\n    498→   holeStunningAmazingCuriosities: hole.stunning_amazing_curiosities,\n    499→   holeAmazingAmazingTrivia: hole.amazing_amazing_trivia,\n    500→   holeIncredibleAmazingTrivia: hole.incredible_amazing_trivia,\n    501→   holeUnbelievableAmazingTrivia: hole.unbelievable_amazing_trivia,\n    502→   holeMindBlowingAmazingTrivia: hole.mind_blowing_amazing_trivia,\n    503→   holeBreathtakingAmazingTrivia: hole.breathtaking_amazing_trivia,\n    504→   holeStunningAmazingTrivia: hole.stunning_amazing_trivia,\n    505→   holeAmazingAmazingFunFacts: hole.amazing_amazing_fun_facts,\n    506→   holeIncredibleAmazingFunFacts: hole.incredible_amazing_fun_facts,\n    507→   holeUnbelievableAmazingFunFacts: hole.unbelievable_amazing_fun_facts,\n    508→   holeMindBlowingAmazingFunFacts: hole.mind_blowing_amazing_fun_facts,\n    509→   holeBreathtakingAmazingFunFacts: hole.breathtaking_amazing_fun_facts,\n    510→   holeStunningAmazingFunFacts: hole.stunning_amazing_fun_facts,\n    511→   holeAmazingAmazingInterestingFacts: hole.amazing_amazing_interesting_facts,\n    512→   holeIncredibleAmazingInterestingFacts: hole.incredible_amazing_interesting_facts,\n    513→   holeUnbelievableAmazingInterestingFacts: hole.unbelievable_amazing_interesting_facts,\n    514→   holeMindBlowingAmazingInterestingFacts: hole.mind_blowing_amazing_interesting_facts,\n    515→   holeBreathtakingAmazingInterestingFacts: hole.breathtaking_amazing_interesting_facts,\n    516→   holeStunningAmazingInterestingFacts: hole.stunning_amazing_interesting_facts,\n    517→   holeAmazingAmazingAmazingFacts: hole.amazing_amazing_amazing_facts,\n    518→   holeIncredibleAmazingAmazingFacts: hole.incredible_amazing_amazing_facts,\n    519→   holeUnbelievableAmazingAmazingFacts: hole.unbelievable_amazing_amazing_facts,\n    520→   holeMindBlowingAmazingAmazingFacts: hole.mind_blowing_amazing_amazing_facts,\n    521→   holeBreathtakingAmazingAmazingFacts: hole.breathtaking_amazing_amazing_facts,\n    522→   holeStunningAmazingAmazingFacts: hole.stunning_amazing_amazing_facts,\n    523→   holeAmazingAmazingAmazingStories: hole.amazing_amazing_amazing_stories,\n    524→   holeIncredibleAmazingAmazingStories: hole.incredible_amazing_amazing_stories,\n    525→   holeUnbelievableAmazingAmazingStories: hole.unbelievable_amazing_amazing_stories,\n    526→   holeMindBlowingAmazingAmazingStories: hole.mind_blowing_amazing_amazing_stories,\n    527→   holeBreathtakingAmazingAmazingStories: hole.breathtaking_amazing_amazing_stories,\n    528→   holeStunningAmazingAmazingStories: hole.stunning_amazing_amazing_stories,\n    529→   holeAmazingAmazingAmazingAnecdotes: hole.amazing_amazing_amazing_anecdotes,\n    530→   holeIncredibleAmazingAmazingAnecdotes: hole.incredible_amazing_amazing_anecdotes,\n    531→   holeUnbelievableAmazingAmazingAnecdotes: hole.unbelievable_amazing_amazing_anecdotes,\n    532→   holeMindBlowingAmazingAmazingAnecdotes: hole.mind_blowing_amazing_amazing_anecdotes,\n    533→   holeBreathtakingAmazingAmazingAnecdotes: hole.breathtaking_amazing_amazing_anecdotes,\n    534→   holeStunningAmazingAmazingAnecdotes: hole.stunning_amazing_amazing_anecdotes,\n    535→   hole
```

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(shotToGreen.shot_number <= par - 2)\n\n// Aggregated:\ngirPercentage = (totalGIR /
totalGIROpportunities) * 100\n// Rounded to 1 decimal: Math.round(value * 1000) /
10\n```\n\n**Data Source:**\n- `golf_rounds.total_gir` (pre-calculated count of GIR holes)\n-
`golf_rounds.total_gir_possible` (total holes played)\n\n**Per-Par
Breakdown:**\n```\ntypescript\ngirPctPar3 = girHitOnPar3 / girOpportunitiesPar3\ngirPctPar4 =
girHitOnPar4 / girOpportunitiesPar4\ngirPctPar5 = girHitOnPar5 /
girOpportunitiesPar5\n```\n\n**GIR by Distance (From which the approach was
hit):**\n```\ntypescript\ngirByDistance = {\n  '50_75': { made: 0, total: 0 },\n  '75_100': {
made: 0, total: 0 },\n  '100_125': { made: 0, total: 0 },\n  // ... continuing to
225+\n}\n\n--\n\n#### Fairways Hit Percentage\n\n**Definition:** Tee shot (par 4s and 5s
only) landed in the fairway\n\n**Calculation:**\n```\ntypescript\nfairwayPercentage =
(fairwaysHit / fairwayOpportunities) * 100\n// Rounded to 1 decimal: Math.round(value * 1000)
/ 10\n\n// Fairways only counted for Par 4 and Par 5s (Par 3s have no fairway)\n```\n\n**Data
Source:**\n- `golf_rounds.total_fairways_hit`\n-
`golf_rounds.total_fairways`\n\n**Breakdowns:**\n```\ntypescript\nfairwayPctPar4 =
fairwaysHitOnPar4 / par4Opportunities\nfairwayPctPar5 = fairwaysHitOnPar5 /
par5Opportunities\nfairwayPctDriver = fairwaysHitWithDriver /
driverAttempts\nfairwayPctNonDriver = fairwaysHitWithNonDriver /
nonDriverAttempts\n```\n\n--\n\n#### Total Putts / Putts Per Round / Putts Per
Hole\n\n**Calculation:**\n```\ntypescript\ntotalPutts = count(shots where shot_type =
'putting')\nnputtsPerRound = totalPutts / roundsPlayed\n// Rounded: Math.round((total /
count) * 100) / 100\n\ninputtsPerHole = totalPutts / totalHolesPlayed\n```\n\n**Data
Source:**\n- Raw shot data: `golf_shots` table filtered by `shot_type = 'putting'` (normalized
to 'putting')\n- Pre-aggregated: `golf_rounds.total_putts`\n\n**Putts Per
GIR:**\n```\ntypescript\nputtsPerGir = totalPutts / girTotal\n// Only counted when GIR was
achieved\n```\n\n--\n\n#### Up-and-Down Percentage (Scrambling)\n\n**Definition:** Missed GIR
but made par or better\n\n**Calculation:**\n```\ntypescript\n// Per hole:\nscrambleAttempt =
!greenInRegulation\nscrambleMade = scrambleAttempt && (score <= par)\n\n//
Aggregated:\nscramblingPercentage = (scramblesMade / scrambleAttempts) * 100\n// Rounded:
Math.round((made / attempts) * 1000) / 10\n```\n\n**Scrambling by Lie (at the chip
position):**\n```\ntypescript\nscramblingPctFairway = fairwayScrambles /
fairwayScrambleAttempts\nscramblingPctRough = roughScrambles /
roughScrambleAttempts\nscramblingPctSand = sandScrambles /
sandScrambleAttempts\n```\n\n**Scrambling by Distance (at the chip position, 0-30
yards):**\n```\ntypescript\nscramblingPct0_10 = scrambles0_10 / attempts0_10\nscramblingPct10_20
= scrambles10_20 / attempts10_20\nscramblingPct20_30 = scrambles20_30 /
attempts20_30\n```\n\n--\n\n#### Sand Save Percentage\n\n**Definition:** Missed GIR in a
bunker but made par or better\n\n**Calculation:**\n```\ntypescript\n// Per
hole:\nsandSaveAttempt = !greenInRegulation && chipLie === 'sand'\nsandSaveMade =
sandSaveAttempt && (score <= par)\n\n// Aggregated:\nsandSavePercentage = (sandSavesMade /
sandSaveAttempts) * 100\n// Rounded: Math.round((made / attempts) * 1000) /
10\n```\n\n**Critical Implementation Note:** Sand saves are tracked using the **chip shot**
(around-green shot) lie, NOT the approach shot lie.\n\n--\n\n### 3. PER-HOLE
STATISTICS\n\n#### Scoring Distribution\n\n**Counts by Score Type:**\n```\ntypescript\n// Per
hole:\nscoreToPar = hole.score - hole.par\n\nif (scoreToPar <= -2) eagles++; \nelse if
(scoreToPar === -1) birdies++; \nelse if (scoreToPar === 0) pars++; \nelse if (scoreToPar === 1)
bogey++; \nelse doublePlus++; \n```\n\n**Per-Round Averages:**\n```\ntypescript\nbirdiesPerRound =
totalBirdies / roundsPlayed\nneaglesPerRound = totalEagles / roundsPlayed\nnparsPerRound =
totalPars / roundsPlayed\nnbogeyPerRound = totalBogeys / roundsPlayed\nndoublePlusPerRound =
totalDoublePlus / roundsPlayed\n```\n\n#### Par-Specific Scoring
Averages\n\n```\ntypescript\npar3Average = (sum of par 3 scores) / (count of par 3
holes)\npar4Average = (sum of par 4 scores) / (count of par 4 holes)\npar5Average = (sum of
par 5 scores) / (count of par 5 holes)\n```\n\n--\n\n### 4. ADVANCED STATISTICS - STROKES
GAINED\n\n**Files:** `~/src/lib/golf/strokes-gained.ts`,
`~/src/lib/utlis/golf-stats-calculator-shots.ts`\n\n#### Formula\n\nStrokes Gained =
Expected Strokes (before) - Expected Strokes (after) - 1\n\n#### Categories\n\n1. **SG:
Off the Tee** - Tee shots on par 4s and 5s\n2. **SG: Approach** - Shots from fairway/rough ≥30
yards intended to reach green\n3. **SG: Around the Green** - Shots <30 yards from green (not

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on green)\n4. **SG: Putting** - All putts on the green\n\n#### PGA Tour Baselines\n\nThe system uses pre-defined PGA Tour benchmark data:\n\n**Off the Tee (Par 5s):**\n\n```\ntypescript\n600: 4.90, 580: 4.85, ... 420: 4.45 yards\n```\n\n**Off the Tee (Par 4s):**\n\n```\ntypescript\n400: 4.10, 380: 4.02, ... 260: 3.62 yards\n```\n\n**Fairway Approach:**\n\n```\ntypescript\n275: 3.45, 250: 3.30, 225: 3.15, ... 20: 2.02 yards\n```\n\n// Mid-range: 50-250 yards\n\n```\n\n**Rough Approach:**\n\n```\ntypescript\n```\n\n// Same distances as fairway but with higher expected strokes\n\n275: 3.65, 250: 3.48, ... (penalty added)\n```\n\n**Sand (Bunker):**\n\n```\ntypescript\n40: 2.65, 35: 2.58, 30: 2.53, ... 3: 2.04 yards\n```\n\n**Putting (Green):**\n\n```\ntypescript\n```\n\n// In FEET\n\n90: 2.40, 80: 2.30, ... 1: 1.00, 0: 0 feet\n```\n\n#### Interpolation\n\nWhen distance doesn't match exact baseline values, **linear interpolation** is used between known data points.\n\n\n#### Data Requirements for Accurate SG Calculation\n\n- ☒ `lie\_before` (starting position)\n- ☒ `distance\_to\_hole\_before` (starting distance)\n- ☒ `lie\_after` (ending position) OR ☒ `result` field\n- ☒ `distance\_to\_hole\_after` (ending distance)\n\n**Returns NULL when incomplete data exists** to avoid skewing averages.\n\n\n---\n\n\n#### 5. PUTTING STATISTICS\n\n\n#### Putting Make Percentage by Distance\n\n```\ntypescript\nputtMakePct0_3 = made0_3 / total0_3 * 100\nputtMakePct3_5 = made3_5 / total3_5 * 100\nputtMakePct5_10 = made5_10 / total5_10 * 100\n```\n\n// ... continuing to 35+\n\n```\n\n**Distance Buckets (Feet):**\n\n- 0-3 feet\n- 3-5 feet\n- 5-10 feet\n- 10-15 feet\n- 15-20 feet\n- 20-25 feet\n- 25-30 feet\n- 30-35 feet\n- 35+ feet\n\n\n#### Putting Statistics by Break Type\n\n```\ntypescript\nputtingByBreak = {\n left_to_right: {\n makePct0_3, makePct3_5, ... , overallMakePct, misses\n },\n straight: {\n ... ,\n right_to_left: {\n ... ,\n multiple: {\n ... }\n }\n }\n }\n}\n```\n\n\n#### Putting Efficiency (Strokes to Hole Out)\n\n```\ntypescript\nputtEff0_5 = average_strokes_to_hole_out_from_0_5_feet\n```\n\n// Examples: 1-putt = 1.0 strokes, 2-putt = 2.0 strokes, etc.\n\n\n#### One-Putt / Three-Putt Counts\n\n```\ntypescript\nnonePuttsTotal = count(holes where putts === 1)\nthreePuttsTotal = count(holes where putts >= 3)\nthreePuttsPerRound = threePuttsTotal / roundsPlayed\n```\n\n\n#### Putting Miss Direction\n\n```\ntypescript\nputtMissLeftPct = missLeft / totalMisses * 100\nputtMissRightPct = missRight / totalMisses * 100\nputtMissShortPct = missShort / totalMisses * 100\nputtMissLongPct = missLong / totalMisses * 100\nputtMissLowPct = missedBreak / totalMisses * 100\nputtMissHighPct = overBreak / totalMisses * 100\n```\n\n// \"Low\" = under-break\n// \"High\" = over-break\n\n\n---\n\n\n#### 6. APPROACH & PROXIMITY STATISTICS\n\n\n#### Approach Proximity (Distance Left After Approach Shot)\n\n```\ntypescript\n```\n\n// ALL approach shots:\n\napproachProximityAvg = avg(distance_to_hole_after)\n\n// When green was hit:\n\napproachProximityWhenHitGreen = avg(proximity | result='green')\n\n// When green was missed:\n\napproachProximityWhenMissedGreen = avg(proximity | result!='green')\n\n// By par:\n\napproachProximityPar3 = avg(proximity | hole.par = 3)\napproachProximityPar4 = avg(proximity | hole.par = 4)\napproachProximityPar5 = avg(proximity | hole.par = 5)\n\n// By lie (from which shot was hit):\n\napproachProximityFairway = avg(proximity | lie_before='fairway')\napproachProximityRough = avg(proximity | lie_before='rough')\napproachProximitySand = avg(proximity | lie_before='sand')\n\n\n#### Approach Proximity by Distance Buckets\n\n```\ntypescript\napproachProx30_75 = avg(proximity | distance_to_hole_before 30-75 yards)\napproachProx75_100 = avg(proximity | distance_to_hole_before 75-100 yards)\n```\n\n// ... continuing to 225+\n\n\n#### Approach Efficiency\n\n```\ntypescript\n```\n\n// Strokes to hole out from approach distance + lie\n\napproachEff30_75 = {\n fairway: avg(strokes_to_hole_out | distance 30-75, lie='fairway'),\n rough: avg(strokes_to_hole_out | distance 30-75, lie='rough'),\n sand: avg(strokes_to_hole_out | distance 30-75, lie='sand')\n}\n\n// Similar for all distance ranges\n\n\n#### Approach Miss Direction (When Not GIR)\n\n```\ntypescript\napproachMissShortPct = missShort / totalApproachMisses * 100\napproachMissLongPct = missLong / totalApproachMisses * 100\napproachMissLeftPct = missLeft / totalApproachMisses * 100\napproachMissRightPct = missRight / totalApproachMisses * 100\napproachMissShortLeftPct = missShortLeft / totalApproachMisses * 100\napproachMissShortRightPct = missShortRight / totalApproachMisses * 100\napproachMissLongLeftPct = missLongLeft / totalApproachMisses * 100\napproachMissLongRightPct = missLongRight / totalApproachMisses * 100\n```\n\n\n---\n\n\n#### 7. DRIVING STATISTICS\n\n\n#### Driving Distance Average\n\n```\ntypescript\ndrivingDistanceAvg = avg(all_driving_distances)\n```\n\n// From tee shots with shot_distance recorded\n\n\ndrivingDistanceDriverOnly = avg(driving_distances | club_type='driver')\n\n// Only

```

when driver was explicitly used\n```\n\n#### Fairway Miss
Direction\n\n```\ntypescript\nmissLeftCount = count(fairway_misses | miss_direction contains
'left')\nmissRightCount = count(fairway_misses | miss_direction contains
'right')\n\nmissLeftPct = missLeftCount / totalMisses * 100\nmissRightPct = missRightCount /
totalMisses * 100\n```\n\n--\n\n\n### 8. AROUND THE GREEN (ATG) STATISTICS\n\n\n#### ATG
Efficiency\n\n```\ntypescript\n// Distance from green after chip/pitch shot\natgEfficiency0_10 =
avg(strokes_to_hole_out | distance 0-10 yards)\natgEfficiency10_20 = avg(strokes_to_hole_out |
distance 10-20 yards)\natgEfficiency20_30 = avg(strokes_to_hole_out | distance 20-30
yards)\n\n// By lie:\natgEffFairway = avg(strokes | lie='fairway')\natgEffRough = avg(strokes
| lie='rough')\natgEffSand = avg(strokes | lie='sand')\n```\n\n\n#### ATG Efficiency by Distance
+ Lie\n\n```\ntypescript\natgEffByDistanceLie = {\n  '0_10': { fairway: null, rough: null, sand:
null },\n  '10_20': { fairway: null, rough: null, sand: null },\n  '20_30': { fairway: null,
rough: null, sand: null }\n}\n\n```\n\n\n--\n\n\n### 9. DATA CACHING & FRESHNESS\n\n\n**Cache
System:**\n- `golf_player_stats_cache` table stores pre-calculated stats\n- Cache marked as
`"stale"` when rounds are edited/deleted\n- RLS policies ensure only authorized users can
access stats\n- Background refresh via `refresh_player_stats_cache()` RPC function\n\n\n**Stale
Cache Handling:**\n\n```\ntypescript\nstats.isStale = boolean flag\n// If true, cache is returned
but refresh triggered in background\n\n```\n\n\n--\n\n\n### 10. TEAM STATISTICS\n\n\nTeam stats
aggregate individual player stats using **raw counts** for proper percentage
calculations:\n\n```\ntypescript\nteamScoringAvg = weighted_average(player_scoring_avg *
rounds_played)\nteamGIR = total_girs / total_gir_opportunities\nteamFairway =
total_fairways_hit / total_fairway_opportunities\n```\n\n\nTeam percentages use **aggregated raw
counts**, NOT averaged percentages, to ensure mathematical accuracy.\n\n\n--\n\n\n### 11.
POTENTIAL ISSUES & EDGE CASES\n\n\n1. **Incomplete Shot Data:** If `distance_to_hole_before` or
`distance_to_hole_after` is NULL, Strokes Gained calculation returns NULL (not 0) to avoid
skewing averages\n\n\n2. **GIR Shot Identification:** For par 5s, if eagle is made earlier than
typical approach position, the eagle shot is used instead of expected approach shot
number\n\n\n3. **Sand Save vs. Scrambling:** Critical distinction:\n  - **Scrambling:** Missed
GIR, made par or better (any lie)\n  - **Sand Save:** Missed GIR in sand specifically, made
par or better\n\n\n4. **Fairway Counting:** Par 3s have no fairway opportunities, so not counted
in fairway percentage\n\n\n5. **Approach Proximity:** Only meaningful when green was actually
hit (distance_to_hole_after is on green), otherwise NULL\n\n\n6. **Rounding Convention:**\n  -
Percentages: `Math.round(value * 1000) / 10` (1 decimal place)\n  - Averages:
`Math.round(value * 100) / 100` (2 decimal places)\n  - Putts: `Math.round(value * 10) / 10`
(1 decimal place)\n\n\n7. **Penalty Strokes:** Tracked separately and excluded from certain SG
categories\n\n\n--\n\n\nThis comprehensive breakdown documents every golf statistic calculation
in the codebase. The system uses a hybrid approach of pre-aggregated round data combined with
detailed shot-level calculations for advanced metrics like Strokes
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TaskOutput

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Show less

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/private/tmp/claude-501/-Users-ricknini-Downloads-helmv3/tasks/a697e0f.output]jils';\n    11→\n    12→type FormatType = 'number' | 'percentage' | 'score' | 'strokes_gained';\n    13→\n    14→interface GolfStatCardProps extends StatCardProps {\n    15→   className?: string;\n    16→   size?: 'sm' | 'md' | 'lg';\n    17→   format?: FormatType;\n    18→   comparisonValue?:
number;\n    19→   comparisonLabel?: string;\n    20→}\n    21→\n    22→export function
GolfStatCard({\n    23→   label,\n    24→   value,\n    25→   trend,\n    26→   trendValue,\n    27→   format = 'number',\n    28→   comparisonValue,\n    29→   comparisonLabel,\n    30→   className,\n    31→   size = 'md',\n    32→}: GolfStatCardProps) {\n    33→   // Format value
based on type\n    34→   const formatDisplayValue = (val: number | string): string => {\n    35→     if (typeof val === 'string') return val;\n    36→\n    37→     switch (format) {\n    38→       case 'percentage':\n    39→         return `${val.toFixed(1)}%`;
    40→       case

```

```

return val > 0 ? `+${val}` : `-${val}`; \n 42→ case
'strokes_gained': \n 43→ return val >= 0 ? `+${val.toFixed(2)}` : val.toFixed(2); \n
44→ default: \n 45→ return Number.isInteger(val) ? val.toString() :
val.toFixed(1); \n 46→ } \n 47→ }; \n 48→ \n 49→ // Get trend icon and color \n
50→ const getTrendStyles = () => { \n 51→ if (!trend) return null; \n 52→ \n 53→
const isPositive = trend === 'improving'; \n 54→ \n 55→ return { \n 56→ icon:
trend === 'improving' ? '↑' : trend === 'declining' ? '↓' : '-', \n 57→ color: trend
=== 'stable' ? 'text-gray-500' : isPositive ? 'text-green-500' : 'text-red-500', \n 58→
bgColor: trend === 'stable' ? 'bg-gray-50' : isPositive ? 'bg-green-50' : 'bg-red-50', \n
59→ }; \n 60→ }; \n 61→ \n 62→ const trendStyles = getTrendStyles(); \n 63→ \n
64→ // Get comparison styles \n 65→ const getComparisonStyles = () => { \n 66→ if
(comparisonValue === undefined || typeof value === 'string') return null; \n 67→ \n 68→
const diff = value - comparisonValue; \n 69→ const isBetter = format === 'strokes_gained'
? diff > 0 : diff < 0; \n 70→ \n 71→ return { \n 72→ diff: Math.abs(diff), \n
73→ isBetter, \n 74→ color: isBetter ? 'text-green-600' : 'text-red-600', \n 75→
}; \n 76→ }; \n 77→ \n 78→ const comparisonStyles = getComparisonStyles(); \n 79→ \n
80→ const sizeStyles = { \n 81→ sm: { \n 82→ container: 'p-3', \n 83→
label: 'text-xs', \n 84→ value: 'text-lg', \n 85→ trend: 'text-xs', \n 86→
}, \n 87→ md: { \n 88→ container: 'p-4', \n 89→ label: 'text-sm', \n 90→
value: 'text-2xl', \n 91→ trend: 'text-xs', \n 92→ }, \n 93→ lg: { \n 94→
container: 'p-6', \n 95→ label: 'text-base', \n 96→ value: 'text-3xl', \n 97→
trend: 'text-sm', \n 98→ }, \n 99→ }; \n 100→ \n 101→ const styles =
sizeStyles[size]; \n 102→ \n 103→ return ( \n 104→ <div \n 105→ className={cn( \n
106→ 'bg-white border border-gray-200 rounded-xl', \n 107→ styles.container, \n
108→ className \n 109→ )} \n 110→ > \n 111→ { /* Label */} \n 112→
<div className={cn('text-gray-500 mb-1', styles.label)}>{label}</div> \n 113→ \n 114→
{ /* Value and Trend */} \n 115→ <div className="flex items-center gap-2"> \n 116→
<span className={cn('font-bold text-gray-900', styles.value)}> \n 117→
{formatDisplayValue(value)} \n 118→ </span> \n 119→ \n 120→ {trendStyles &&
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* MissPatternChart - Circular/Radar Visualization for Miss Patterns \n 5→ * \n 6→ *
Premium visualization showing miss directions as segments in a circular chart. \n 7→ * Uses
custom SVG for a distinctive, non-generic appearance. \n 8→ */ \n 9→ \n 10→import {
useMemo } from 'react'; \n 11→import { motion } from 'framer-motion'; \n 12→import { cn }
from '@/lib/utills'; \n 13→ \n 14→interface MissPatternData { \n 15→ direction:

```



```

string;\n    16→ percentage: number;\n    17→ count?: number;\n    18→}\n    19→\n
20→interface MissPatternChartProps {\n    21→ data: MissPatternData[];\n    22→ title?:
string;\n    23→ subtitle?: string;\n    24→ size?: 'sm' | 'md' | 'lg';\n    25→
className?: string;\n    26→ showLegend?: boolean;\n    27→ animated?: boolean;\n    28→}\n
29→\n    30→// Direction positions for the circular chart (in degrees from top, clockwise)\n
31→const DIRECTION_ANGLES: Record<string, number> = {\n    32→ short: 180,\n    33→ long:
0,\n    34→ left: 270,\n    35→ right: 90,\n    36→ short_left: 225,\n    37→ short_right:
135,\n    38→ long_left: 315,\n    39→ long_right: 45,\n    40→};\n    41→\n    42→//
Friendly labels for directions\n    43→const DIRECTION_LABELS: Record<string, string> = {\n
44→ short: 'Short',\n    45→ long: 'Long',\n    46→ left: 'Left',\n    47→ right:
'Right',\n    48→ short_left: 'Short-Left',\n    49→ short_right: 'Short-Right',\n    50→
long_left: 'Long-Left',\n    51→ long_right: 'Long-Right',\n    52→};\n    53→\n    54→//
Colors based on severity/frequency\n    55→function getSegmentColor(percentage: number):
string {\n    56→ if (percentage >= 50) return '#ef4444'; // Red - critical\n    57→ if
(percenta
ge >= 35) return '#f59e0b'; // Amber - warning\n    58→ if (percentage >= 20) return
'#22c55e'; // Green - moderate\n    59→ return '#94a3b8'; // Slate - low\n    60→}\n    61→\n
62→const sizes = {\n    63→ sm: { viewBox: 200, center: 100, radius: 75, labelRadius: 95 },\n
64→ md: { viewBox: 280, center: 140, radius: 105, labelRadius: 130 },\n    65→ lg: {
viewBox: 360, center: 180, radius: 140, labelRadius: 170 },\n    66→};\n    67→\n    68→export
function MissPatternChart({\n    69→ data,\n    70→ title = 'Miss Pattern',\n    71→
subtitle,\n    72→ size = 'md',\n    73→ className,\n    74→ showLegend = true,\n    75→
animated = true,\n    76→}: MissPatternChartProps) {\n    77→ const { viewBox, center,
radius, labelRadius } = sizes[size];\n    78→\n    79→ // Sort data by percentage for
consistent rendering\n    80→ const sortedData = useMemo(() => {\n    81→ return
[...data]\n    82→ .filter(d => d.percentage > 0)\n    83→ .sort((a, b) =>
b.percentage - a.percentage);\n    84→ }, [data]);\n    85→\n    86→ // Calculate segment
arc paths\n    87→ const segments = useMemo(() => {\n    88→ return sortedData.map((item)
=> {\n    89→ const angleDeg = DIRECTION_ANGLES[item.direction] ?? 0;\n    90→ const
angleRad = (angleDeg - 90) * (Math.PI / 180);\n    91→\n    92→ // Segment arc span (45
degrees for 8 directions)\n    93→ const arcSpan = 40;\n    94→ const startAngle =
angleDeg - arcSpan / 2;\n    95→ const endAngle = angleDeg + arcSpan / 2;\n    96→\n
97→ // Calculate segment radius based on percentage\n    98→ const segmentRadius =
(radius * 0.4) + (radius * 0.6 * (item.percentage / 100));\n    99→\n    100→ // Arc
path\n    101→ const startRad = (startAngle - 90) * (Math.PI / 180);\n    102→ const
endRad = (endAngle - 90) * (Math.PI / 180);\n    103→\n    104→ const x1 = center +
segmentRadius * Math.cos(startRad);\n    105→ const y1 = center + segmentRadius *
Math.sin(startRad);\n    106→ const x2 = center + segmentRadius * Math.cos(endRad);\n
107→ const y2 = center + segmentRadius * Math.sin(endRad);\n    108→\n    109→ const
largeArc = arcSpan > 180 ? 1 : 0;\n    110→\n    111→ const path = `\n    112→ M
${center} ${center}\n    113→ L ${x1} ${y1}\n    114→ A ${segmentRadius}
${segmentRadius} 0 ${largeArc} 1 ${x2} ${y2}\n    115→ Z\n    116→ `;\n    117→\n
118→ // Label position\n    119→ const labelX = center + labelRadius *
Math.cos(angleRad);\n    120→ const labelY = center + labelRadius *
Math.sin(angleRad);\n    121→\n    122→ return (\n    123→ <div\n    124→ <div>
{title}</div>\n    125→ <div>{subtitle}</div>\n    126→ <div>\n    127→ <div>
{path}</div>\n    128→ <div>{labelX}</div> {labelY}</div>\n    129→ </div>\n    130→
)};\n    131→\n    132→\n    133→\n    134→\n    135→\n    136→\n    137→\n    138→\n
139→\n    140→\n    141→\n    142→\n    143→\n    144→\n    145→\n    146→\n    147→\n
148→\n    149→\n    150→\n    151→\n    152→\n    153→\n    154→\n    155→\n    156→\n
157→\n    158→\n    159→\n    160→\n    161→\n    162→\n    163→\n    164→\n    165→\n
166→\n    167→\n    168→\n    169→\n    170→\n    171→\n    172→\n    173→\n    174→\n
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193→\n    194→\n    195→\n    196→\n    197→\n    198→\n    199→\n    200→\n    201→\n
202→\n    203→\n    204→\n    205→\n    206→\n    207→\n    208→\n    209→\n    210→\n
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Selection** - loads detailed stats on demand (lazy loading)\n4. **Caching Strategy** -
`detailedStatsCache` useRef prevents re-fetching same data\n\n**Key
Queries:**\n```\ntypescript\n// Load team players\n.from('golf_teams')\n .select('id')\n
.eq('organization_id', organizationId)\n\n.from('golf_team_members')\n .select('player_id')\n
.eq('team_id', teamId)\n .eq('status', 'active')\n\n// Load rounds with shot
data\n.from('golf_rounds')\n .select(`\n id, round_date, course_name, round_type,\n
total_score, score_to_par, total_fairways_hit,\n      total_fairways, total_gir,
total_gir_possible, total_putts\n `)\n```\n\n---\n\n### 2. GOLF STATS DISPLAY
COMPONENT\n\n**Location:**
`/src/components/golf/stats/GolfStatsDisplay.tsx`\n\n**Architecture:**\n- Complex animation
system with Framer Motion\n- Multiple custom hooks for animations (useAnimatedNumber)\n-
Tab-based interface for different stat categories\n- Dynamic component rendering with
motion.AnimatePresence\n\n**Key Features:**\n- **Animated Numbers** - Smooth transition of
stat values with ease-out cubic function\n- **Floating Controls** - Mobile nav, back button,
refresh button with animations\n- **Round Selector** - Toggle between 'overall' and individual
round stats\n- **Filter System** - Dynamic filtering by course, round type, date range\n-
**Player Profile Header** - Shows avatar, grad year, handicap (coach view only)\n\n**Rendered
Stats Sections:**\n```\n1. Overview Tab - Summary KPIs\n2. Scoring Tab - Score distribution,
scoring average\n3. Driving Tab - Fairway %, miss directions, distance\n4. Approach Tab - GIR
%, miss patterns, lie analysis\n5. Putting Tab - Putt metrics, miss zones, 3-putt tracking\n6.
Scrambling Tab - Up & down statistics\n7. Strokes Gained Tab - SG breakdown by category\n8.
Progress Tab - Trend lines, round history\n```\n\n---\n\n### 3. SPRAY CHARTS & SHOT
VISUALIZATIONS\n\n**Premium Dispersion Components** (Foundation-First Design):\n\n#### A.
**DrivingDispersionPremium.tsx**\n- **Purpose:** Visualize tee shot accuracy
(left/fairway/right)\n- **Visual:** SVG fairway with 3 zones using perspective transform\n-
**Data Points:**\n - `fairwayPct` - Primary stat (dominant display)\n - `missLeftCount` /
`missRightCount` - Rough miss patterns\n - Shot dots animated in with spring physics\n-
**Zones:**\n - Left rough: Red gradient (#fecaca to #fca5a5) - opacity 0.2-0.6\n - Fairway:
Green gradient (#22c55e to #16a34a) - opacity 0.25-0.5\n - Right rough: Orange gradient
(#fed7aa to #fdb462) - opacity 0.2-0.6\n- **Hover States:** Zone highlights on mouseover, shot
dots enlarge\n- **SVG Dimensions:** 240x140 viewBox with perspective path for realism\n\n####
B. **ApproachDispersionPremium.tsx**\n- **Purpose:** Show approach shot accuracy from
different lies\n- **Visual:** Circular green visualization with GIR shots + miss directions\n-
**Data Points:**\n - `girPct` - Green hit percentage\n - `girFromFairway` / `girFromRough` /
`girFromSand` - Lie breakdowns\n - Miss directions: short/long/left/right with percentages\n-
**Shot Positioning:**\n - GIR shots: Circular arrangement around green center\n - Misses:
Radial positions based on direction (short=bottom, long=top)\n- **Color Coding:**\n - GIR:
Green dots arranged in circle\n - Short/Long/Left/Right: Distinct colors (#3b82f6, #8b5cf6,
#ef4444, #f97316)\n\n#### C. **PuttingDispersionPremium.tsx**\n- **Purpose:** Visualize
putting miss patterns (left/right/short/long)\n- **Visual:** Green rectangle with 4 miss zones
+ center hole\n- **Data Points:**\n - `puttMissLeftPct`, `puttMissRightPct`,
`puttMissShortPct`, `puttMissLongPct`\n - `onePuttsTotal`, `threePuttsTotal` - 1-putt and
3-putt counts\n - `totalPutts` - Total putts for the round/period\n- **Miss Zones:** 4
cardinal directions (left/right/short/long)\n- **SVG Grid:** 200x200 viewBox with miss dot
clustering\n- **Color Legend:**\n - Red (left/right misses)\n - Blue (short misses)\n -
Purple (long misses)\n\n#### D. **ShotDispersionChart.tsx**\n- **Purpose:** Master container
for all 3 dispersion charts\n- **Wrapper:** Coordinates the three premium dispersion
components\n- **Animation:** Staggered entrance with shot dot animations\n- **Props:** Passes
all shot data down to sub-components\n\n---\n\n### 4. CHART COMPONENTS & TREND
VISUALIZATIONS\n\n#### A. **TrendLineChart.tsx**\n- **Type:** Line chart with area fill +
moving average\n- **Features:**\n - Multi-line support (main data + 5-point moving average)\n
- Prediction visualization (dashed line extending forward)\n - Interactive hoverable data
points\n - Y-axis grid lines and labels\n - X-axis date labels (smart spacing to prevent
crowding)\n- **Data Format:**\n```\ntypescript\nTrendPoint = { date: string; value: number
}\nTrendAnalysis = { direction: 'improving'|'declining'|'stable'; changePercent: number }\n```\n-
**Visualization:**\n - Main line: 2.5px stroke with rounded caps\n - Area fill: 10%
opacity colored fill below line\n - Moving average: Dashed line (4,4 pattern) in slate
color\n - Prediction: Dashed to future point with circular marker\n- **Dimensions:** 600px

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width, configurable height (default 200px)\n\n#### B. ****StrokesGainedRadar.tsx****\n- ****Type:**** Radar/Spider chart for strokes gained\n- ****Categories:****\n - Off the Tee (SG_off_tee)\n - Approach (SG_approach)\n - Around Green (SG_around_green)\n - Putting (SG_putting)\n- ****Visualization:****\n - 4 axes positioned at compass points\n - Grid circles at [-1, 0, 1, 2] SG levels\n - Player polygon filled with semi-transparent green\n - Optional comparison polygon overlay\n - Labels at endpoints for each category\n- ****Scale:**** 2 SG = full radius (scaleFactor = radius/2)\n- ****Coloring:****\n - Player: Green polygon (#22c55e)\n - Comparison: Alternative color with reduced opacity\n - Grid lines: Slate color (#94a3b8) for zero line, lighter for others\n\n#### C. ****StrokesGainedDashboard.tsx****\n- ****Container:**** Combines radar + breakdown charts + trends\n- ****Components:****\n - StrokesGainedRadar (primary overview)\n - StrokesGainedBreakdownChart (category bars)\n - TrendLineChart (for each SG category over time)\n - Comparison data display\n- ****Tabs:**** Overview, Breakdown, Trends, Insights\n- ****Features:****\n - Identifies strengths/weaknesses automatically\n - Shows primary strength and weakness highlighted\n - Badge display for key insights\n\n#### D. ****PerformanceHeatmap.tsx****\n- ****Type:**** GitHub-style calendar heatmap for round history\n- ****Purpose:**** Show distribution of rounds and scores by day/week\n- ****Features:****\n - Year selector (navigate to previous/next year)\n - Calendar grid with color-coded round cells\n - Hover to see round details\n - Click handler for selecting specific round\n- ****Color Scheme:****\n - Emerald 600: Excellent rounds (≤ -5 to par)\n - Emerald 400: Good rounds (< 0 to par)\n - Slate 300: Even par\n - Amber 400: Over par by 1-3\n - Orange 400: Over par by 4-6\n - Red 500: Bad rounds (> 6 over par)\n- ****Statistics:**** Total rounds, avg score, best round, round distributions\n- ****Calendar Layout:**** ISO date positioning with Sunday start\n\n#### E. ****TrendVisualization.tsx****\n- ****Purpose:**** Multi-metric trend comparison\n- ****Features:**** Dashboard showing trend directions for multiple stats\n- ****Displays:**** Scoring average, putts, fairway %, GIR %, strokes gained categories\n- ****Indicators:**** Up/down/stable arrows with color coding\n\n---\n\n#### 5. COACHHELM ANALYTICS COMPONENTS\n\n#### A. ****ShotAnalyticsPanel.tsx****\n- ****Master Component:**** Combines all shot analytics\n- ****Tabs:****\n - Overview - Summary statistics\n - Patterns - Miss pattern visualizations\n - Breakdown - Shot type distribution\n - Trends - Temporal trends\n- ****Controls:****\n - Period selector (7/14/30/60/90 days)\n - Refresh button with loading state\n - Error display capability\n- ****Data Source:**** ``getPlayerShotAnalytics()`` server action\n\n#### B. ****ShotTypeBreakdown.tsx****\n- ****Purpose:**** Show distribution of shot types\n- ****Format:**** Horizontal bar chart or pie chart\n- ****Categories:**** Drive, Approach, Putt, Chip, etc.\n\n#### C. ****MissPatternChart.tsx****\n- ****Type:**** Circular/radial segments for miss directions\n- ****Positioning:**** 8-direction layout (short, long, left, right, diagonal combos)\n- ****Segment Sizing:**** Radius grows with percentage\n- ****Color:**** Based on severity\n - Red ($\geq 50\%$)\n - Amber ($\geq 35\%$)\n - Green ($\geq 20\%$)\n - Slate ($< 20\%$)\n- ****Interactivity:**** Hover highlights direction\n\n#### D. ****TrendIndicator.tsx****\n- ****Purpose:**** Show metric improvement/decline\n- ****Display:**** Icon ($\uparrow$, $\rightarrow$) + label + optional metadata\n\n#### E. ****PatternImpactPanel.tsx****\n- ****Purpose:**** Show how patterns affect overall performance\n- ****Analysis:**** Connects miss patterns to scoring impact\n\n#### F. ****AnalyticsSummaryCards.tsx****\n- ****Layout:**** Grid of KPI cards\n- ****Content:**** Key metrics at a glance\n\n#### G. ****InsightEffectivenessPanel.tsx****\n- ****Purpose:**** Show effectiveness of coaching insights\n- ****Metrics:**** Improvement before/after coaching focus\n\n#### H. ****PredictionAccuracyPanel.tsx****\n- ****Purpose:**** AI prediction accuracy metrics\n- ****Display:**** Calibration chart or success rate\n\n---\n\n#### 6. STATS DATA CALCULATION\n- ****Location:**** ``/src/lib/utils/golf-stats-calculator-shots.ts``\n- ****Single Source of Truth:**** ``golf_shots`` table (raw shot data)\n- ****Output Type:**** `GolfStats`\n\n

```
\ninterface GolfStats {\n  // General\n  roundsPlayed: number;\n  holesPlayed: number;\n\n  // Scoring\n  scoringAverage: number | null;\n  avgScoreToPar: number | null;\n  bestRound: number | null;\n  worstRound: number | null;\n  totalBirdies: number;\n  totalEagles: number;\n  totalPars: number;\n  totalBogeys: number;\n  totalDoublePlus: number;\n  birdiesPerRound: number | null;\n\n  // Driving (140+ stats)\n  drivingDistanceAvg: number | null;\n  fairwaysHit: number;\n  fairwayPercentage: number | null;\n  missLeftCount: number;\n  missRightCount: number;\n  missLeftPct: number | null;\n  missRightPct: number | null;\n\n  // GIR\n  girTotal: number;\n  girPercentage: number | null;\n  girPctFromFairway: number | null;\n  girPctFromRough: number | null;\n  girPctFromSand: number | null;\n\n  // Approach Miss Directions\n  approachMissShortPct:
```

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number | null;\n  approachMissLongPct: number | null;\n  approachMissLeftPct: number | null;\n
approachMissRightPct: number | null;\n\n  // Putting\n  puttMissLeftPct: number | null;\n
puttMissRightPct: number | null;\n  puttMissShortPct: number | null;\n  puttMissLongPct:
number | null;\n  puttsPerRound: number | null;\n  onePuttPercentage: number | null;\n
threePuttPercentage: number | null;\n\n  // Strokes Gained (requires baseline data)\n
sgTotal: number | null;\n  sgOffTee: number | null;\n  sgApproach: number | null;\n
sgAroundGreen: number | null;\n  sgPutting: number | null;\n\n  // Scrambling\n
scramblingTotal: number;\n  scramblingPercentage: number | null;\n)\n\n```\n\n**Server Action:**
`/src/app/golf/actions/stats-data.ts`\n\n```\ntypescript\n\nexport async function
getDetailedStats(\n  playerId: string,\n  roundId: string | 'overall' = 'overall',\n  filter?:
StatsFilter\n): Promise<GolfStats>\n\n\nexport async function getFilterOptions(playerId:
string): Promise<FilterOptions>\n\n\nexport async function getCourseBreakdown(playerId: string):
Promise<CourseBreakdownResponse>\n\n\nexport async function getWorstHoleAnalysis(playerId:
string): Promise<WorstHoleResponse>\n\n\nexport async function getTrendAnalysis(playerId:
string): Promise<TrendAnalysisResponse>\n\n\n```\n\n**Filter Types:**\n\n```\ntypescript\n\ninterface
StatsFilter {\n  preset?: 'last5' | 'last10' | 'last20' | 'tournaments' | 'practice' |
'thisMonth' | 'thisYear';\n  startDate?: string;\n  endDate?: string;\n  courseName?:
string;\n  roundType?: 'practice' | 'qualifier' | 'tournament';\n  season?:
number;\n}\n\n```\n\n---\n\n\n### 7. COACH VS PLAYER STATS VIEWS\n\n\n**Coach View
(isCoachView={true}):**\n- **Roster List:** All team players with rank, avatar, sparkline,
stats block\n- **Sort Options:** By avg score, name, improvement trend, recent activity\n-
**Filters:** Class year, search by name\n- **Player Selection:** Click card to drill into
individual stats\n- **KPI Summary:** Team average, total rounds, top 5 average, improving
count\n- **Player Profile Header:** Avatar, grad year, handicap, rounds played, scoring avg,
best round\n\n\n**Player View (isCoachView={false}):**\n- **Self Stats:** Own stats displayed
prominently\n- **No Roster:** Single player context\n- **Profile Comparison:** Can't see other
players' detailed stats\n- **Same Stat Categories:** All stat tabs available\n\n\n**Shared UI
Elements:**\n- Identical stat calculation method\n- Same chart components\n- Same
filtering/round selection\n- Mobile navigation toggle\n- Floating action buttons\n\n\n---\n\n\n###
8. KEY COMPONENTS SUMMARY\n\n\n| Component | Purpose | Type | Key Props |\n|-----|-----|-----|-----|\n| GolfStatsDisplay | Main stats container |
Wrapper | stats, playerName, isCoachView, rounds |\n| DrivingDispersionPremium | Fairway
visualization | SVG Chart | fairwayPct, missLeftCount, missRightCount |\n| ApproachDispersionPremium | Green accuracy | SVG Chart | girPct, missShort/Long/Left/RightPct |\n| PuttingDispersionPremium | Putting misses | SVG Chart | puttMissLeft/Right/Short/LongPct |\n| ShotDispersionChart | Master dispersion | Container | Complete shot data |\n| TrendLineChart | Line graph trends | Recharts-based | data, analysis, metric, color |\n| StrokesGainedRadar | SG overview | SVG Radar | sgOffTee, sgApproach, sgAroundGreen, sgPutting |\n| PerformanceHeatmap | Calendar heatmap | SVG Grid | rounds, year, onRoundClick |\n| ProgressStats | Score trends + rounds | Recharts | stats, rounds |\n| ShotAnalyticsPanel | CoachHelm analytics | Container | playerId, playerName, periodDays |\n| MissPatternChart | Circular miss viz | SVG Radial | data[], title, size |\n\n\n---\n\n\n### 9. DATA FLOW
DIAGRAM\n\n\n```\nUSER (Coach or Player)\n  ↓\nstats/page.tsx (Route wrapper)\n  ↓\nStatsClient (Main logic, 1,070 lines)\n  ├── Load user role (coach vs player)\n  ├── Load coach: All team players\n  │   ├── Load player: Own data only\n  │   └── Lazy load detailed stats on tab click\n  │       └── Cache with useRef Map\n  └── ↓\n      GolfStatsDisplay (Tab container)\n        ├── StatTab: Overview\n        │   ├── StatTab: Scoring\n        │   └── StatTab: Driving → DrivingDispersionPremium\n        ├── StatTab: Approach → ApproachDispersionPremium\n        ├── StatTab: Putting → PuttingDispersionPremium\n        ├── StatTab: Scrambling\n        └── StatTab: Strokes Gained → StrokesGainedDashboard\n            ├── StatTab: Progress → ProgressStats\n            └── ↓\n                ShotDispersionChart (Combines all 3 dispersion charts)\n                  ├── DrivingDispersionPremium\n                  ├── ApproachDispersionPremium\n                  └── PuttingDispersionPremium\n                      ↓\nDatabase: golf_shots (Single source of truth)\n  → Calculate GolfStats\n  → Render visualizations\n\n\n---\n\n\n### 10. ANIMATION & STYLING\n\n\n**Framer Motion Patterns:**\n- `containerVariants` - Stagger child animations (0.05s delay)\n- `itemVariants` - Spring physics (stiffness: 300, damping: 24)\n- `shotDotVariants` - Individual shot dots with custom delay\n- `tabContentVariants` - Slide in/out on tab change (x: 20px offset)\n\n\n**Color System:**\n- **Primary:** #16A34A (Kelly Green)\n- **Cream Background:** #FFFEFA\n- **Text:**

```



```

`gpa` (DECIMAL(3,2))\n- `status` (golf_player_status ENUM: 'active', 'injured', 'redshirt', 'inactive')\n- `onboarding_completed`, `profile_complete` (BOOLEAN)\n- `created_at`, `updated_at` (TIMESTAMP)\n\n**Indexes**:\n- `idx_golf_players_user` (user_id)\n- `idx_golf_players_team` (team_id)\n- `idx_golf_players_status` (status)\n- `idx_golf_players_user_status` (user_id, status) - added in Migration 050\n\n**RLS**:
```

Not explicitly enabled (relies on team/coach relationships)\n\n---\n\n### 1.2 golf_rounds (Primary Stats Table)\n\n**Location**:

```

Migration 021_golf_rounds.sql\n\n**Primary Key**:
```

UUID\n\n**Key Columns - Core Data**:\n- `id` (UUID, primary key)\n- `player\_id` (UUID, FK → golf_players, ON DELETE CASCADE) ★\n- `qualifier\_id` (UUID, FK added after golf_qualifiers created)\n- `course\_id` (UUID, FK → golf_courses, ON DELETE SET NULL) ★\n- `round\_number` (INTEGER)\n- `course\_name`, `course\_city`, `course\_state` (TEXT)\n- `course\_rating` (DECIMAL(4,1))\n- `course\_slope` (INTEGER)\n- `tees\_played` (TEXT)\n- `round\_date` (DATE) ★\n- `round\_type` (golf_round_type ENUM: 'tournament', 'qualifier', 'practice', 'casual') - Migration 001\n- `round\_status` (golf_round_status ENUM: 'in_progress', 'paused', 'completed', 'abandoned') - Migration 001\n\n**Scoring Columns**:\n- `total\_score` (INTEGER) - Sum of hole scores\n- `total\_to\_par` (INTEGER) - Score - Total Par\n- `total\_putts` (INTEGER)\n- `fairways\_hit`, `fairways\_total` (INTEGER)\n- `greens\_in\_regulation`, `greens\_total` (INTEGER)\n- `total\_penalties` (INTEGER)\n\n**Metadata**:\n- `notes` (TEXT)\n- `is\_verified` (BOOLEAN)\n- `verified\_by` (UUID, FK → golf_coaches)\n- `verified\_at` (TIMESTAMP)\n- `weather\_conditions`, `wind\_speed` (TEXT)\n- `temperature` (INTEGER)\n- `created\_at`, `updated\_at` (TIMESTAMP)\n\n**Indexes**:

```

Migration 050:\n- `idx_golf_rounds_player` (player_id)\n- `idx_golf_rounds_date` (round_date DESC)\n- `idx_golf_rounds_qualifier` (qualifier_id) WHERE qualifier_id IS NOT NULL\n- `idx_golf_rounds_type` (round_type)\n- `idx_golf_rounds_status` (round_status)\n- `idx_golf_rounds_player_status_date` (player_id, status, round_date DESC) - COMPOSITE for stats queries\n- `idx_golf_rounds_date_player` (round_date DESC, player_id) - For coach view\n- `idx_golf_rounds_in_progress` (player_id, updated_at DESC) WHERE status = 'in_progress' - PARTIAL\n\n**Helper Function** (Migration 021):\n- `sql\ncalculate_round_stats(p_round_id UUID)`\n- Recalculates total_score, total_to_par, total_putts, fairways, GIR, penalties from golf_holes\n\n**RLS**:
```

Not explicitly enabled\n\n---\n\n### 1.3 golf_holes (Individual Hole Scores)\n\n**Location**:

```

Migration 021_golf_rounds.sql\n\n**Primary Key**:
```

UUID\n\n**Key Columns**:\n- `id` (UUID, primary key)\n- `round\_id` (UUID, FK → golf_rounds, ON DELETE CASCADE) ★\n- `hole\_number` (INTEGER, 1-18)\n- `par` (INTEGER, 3-5)\n- `yardage` (INTEGER)\n- `score` (INTEGER, ≥ 1)\n- `score\_to\_par` (INTEGER GENERATED ALWAYS AS (score - par) STORED)\n- `putts` (INTEGER, ≥ 0)\n- `fairway\_hit`, `green\_in\_regulation`, `up\_and\_down`, `sand\_save` (BOOLEAN)\n- `penalties` (INTEGER, default 0)\n- `notes` (TEXT)\n\n**Constraints**:\n- UNIQUE(round_id, hole_number) - Ensures one hole score per round\n\n**Indexes**:\n- `idx\_golf\_holes\_round` (round_id)\n- `idx\_golf\_holes\_round\_number` (round_id, hole_number) - COMPOSITE, Migration 050\n\n**RLS**:

Not explicitly enabled\n\n---\n\n### 1.4 golf_shots (Shot-by-Shot Tracking)\n\n**Location**:

```

Migration 021_golf_rounds.sql (core), enhanced by Migration 040, 064-066\n\n**Primary Key**:
```

UUID\n\n**Key Columns - Core**:\n- `id` (UUID, primary key)\n- `hole\_id` (UUID, FK → golf_holes, ON DELETE CASCADE) ★\n- `shot\_number` (INTEGER)\n- `club` (golf_shot_club ENUM: 'driver', '3_wood', '5_wood', '7_wood', 'hybrid', '2_iron'-'9_iron', 'pw', 'gw', 'sw', 'lw', 'putter') - Migration 001\n- `distance\_yards` (INTEGER)\n- `result` (golf_shot_result ENUM: 'excellent', 'good', 'average', 'poor', 'miss') - Migration 001\n- `miss\_direction` (golf_miss_direction ENUM: 'left', 'right', 'short', 'long', 'left_short', 'left_long', 'right_short', 'right_long') - Migration 001\n- `lie` (TEXT: 'fairway', 'rough', 'bunker', etc.)\n- `notes` (TEXT)\n\n**Enhanced Columns (Migration 040)**:\n- `club\_type` (TEXT: 'driver', 'non_driver', 'putter') - For strokes gained benchmarks\n- `shot\_type` (TEXT: 'tee', 'approach', 'around_green', 'putting', 'penalty') - CHECK constraint\n- `lie\_before` (TEXT: 'tee', 'fairway', 'rough', 'sand', 'green', 'other')\n- `distance\_unit\_before`, `distance\_unit\_after` (TEXT: 'yards', 'feet')\n- `distance\_to\_hole\_before`, `distance\_to\_hole\_after` (NUMERIC(7,2)) - CRITICAL for approach stats! (Migration 066)\n- `shot\_distance` (NUMERIC(7,2)) - Derived: before - after\n- `lie\_after` (TEXT: 'tee', 'fairway', 'rough', 'sand', 'green', 'penalty', 'recovery', 'other')\n- `is\_penalty` (BOOLEAN)\n- `penalty\_type` (TEXT: 'ob', 'water', 'unplayable', 'lost')\n\n**Putting-Specific (Migration 065)**:\n- `putt\_distance\_feet` (NUMERIC(5,1)) - Distance of putt\n- `putt\_made` (BOOLEAN) - Whether putt was holed\n- `putt\_break` (TEXT: 'left_to_right', 'right_to_left',


```

'straight', 'multiple'))\n- `putt_slope` (INTEGER) - Estimated slope
percentage\n\n**Timestamps**:\n- `created_at`, `updated_at` (TIMESTAMPTZ)\n\n**Indexes** -
Migration 040, 050, 065, 066:\n- `idx_golf_shots_hole` (hole_id)\n-
`idx_golf_shots_round_hole_shot` (round_id, hole_number, shot_number) - COMPOSITE\n-
`idx_golf_shots_round_type` (round_id, shot_type) - For SG calculations\n-
`idx_golf_shots_round_id` (round_id) - For bulk deletions\n- `idx_golf_shots_shot_type`
(shot_type)\n- `idx_golf_shots_lie_before` (lie_before)\n- `idx_golf_shots_result` (result)\n-
`idx_golf_shots_putt_made` (putt_made) WHERE putt_made IS NOT NULL\n-
`idx_golf_shots_lie_after` (lie_after) WHERE lie_after IS NOT NULL\n-
`idx_golf_shots_putting_analysis` (shot_type, putt_made, putt_distance_feet) WHERE shot_type =
'putting'\n- `idx_golf_shots_distance_before` (distance_to_hole_before) WHERE
distance_to_hole_before IS NOT NULL\n\n**CHECK Constraints** (Migration 040):\n``\n-
shot_type IN ('tee', 'approach', 'around_green', 'putting', 'penalty')\n- club_type IN
('driver', 'non_driver', 'putter')\n- lie_before IN ('tee', 'fairway', 'rough', 'sand',
'green', 'other')\n- result IN ('fairway', 'rough', 'sand', 'green', 'hole', 'other',
'penalty')\n- putt_break IN ('left_to_right', 'right_to_left', 'straight', 'multiple')\n-
distance_unit_before IN ('yards', 'feet')\n- distance_unit_after IN ('yards', 'feet')\n-
lie_after IN ('tee', 'fairway', 'rough', 'sand', 'green', 'penalty', 'recovery',
'other')\n``\n\n**RLS**:\n- Not explicitly enabled in golf_shots\n\n--\n\n## 2. STATS CACHE
TABLES (Pre-calculated Performance Data)\n\n### 2.1 golf_player_stats_cache\n\n**Location**:\n
Migration 040_golf_shot_system.sql\n\n**Purpose**:\n- Cache expensive multi-round calculations for
instant dashboard loads\n\n**Primary Key**:\n- UUID\n\n**Unique Constraint**:\n- player_id (one cache
record per player)\n\n**FK**:\n- player_id → golf_players(id) ON DELETE CASCADE\n\n**Scoring
Aggregate**:\n- `scoring_average` (NUMERIC(5,2))\n- `scoring_average_vs_par` (NUMERIC(5,2))\n-
`rounds_played` (INTEGER, default 0)\n- `best_round`, `worst_round` (INTEGER)\n\n**Par
Performance Breakdown**:\n- `par3_average`, `par4_average`, `par5_average`
(NUMERIC(5,2))\n\n**Scoring Distribution**:\n- `eagles`, `birdies`, `pars`, `bogey`,
`double_bogey`, `triple_plus` (INTEGER, defaults 0)\n\n**Strokes Gained (PGA Tour
Benchmarks)**:\n- `strokes_gained_total`, `strokes_gained_tee`, `strokes_gained_approach`,
`strokes_gained_around_green`, `strokes_gained_putting` (NUMERIC(5,2))\n\n**Driving**:\n-
`driving_accuracy_percentage` (NUMERIC(5,2))\n- `fairways_hit`, `fairways_total` (INTEGER,
defaults 0)\n- `driving_distance_average` (NUMERIC(6,1))\n\n**Approach & GIR**:\n-
`gir_percentage` (NUMERIC(5,2))\n- `greens_hit`, `greens_total` (INTEGER, defaults 0)\n-
`approach_proximity_average` (NUMERIC(6,1)) - Distance to hole on approach shots\n\n**Short
Game**:\n- `scrambling_percentage`, `scrambles_converted`, `scramble_attempts` (NUMERIC(5,2) /
INTEGER)\n- `sand_save_percentage`, `sand_saves`, `sand_attempts` (NUMERIC(5,2) / INTEGER)\n-
`up_and_down_percentage` (NUMERIC(5,2))\n\n**Putting Overall**:\n- `putts_per_round`,
`putts_per_gir` (NUMERIC(4,2))\n- `one_putt_percentage`, `three_putt_percentage`
(NUMERIC(5,2))\n- `total_putts` (INTEGER, default 0)\n\n**Putting by Distance (feet)**:\n-
`putt_make_pct_0_3ft`, `putt_make_pct_3_5ft`, `putt_make_pct_5_10ft`, `putt_make_pct_10_15ft`,
`putt_make_pct_15_20ft`, `putt_make_pct_20_plus_ft` (NUMERIC(5,2))\n\n**Putting by Break
Type**:\n- `putt_make_pct_left_to_right`, `putt_make_pct_right_to_left`,
`putt_make_pct_straight` (NUMERIC(5,2))\n\n**Penalties**:\n- `penalty_strokes_per_round`
(NUMERIC(4,2))\n- `total_penalties` (INTEGER, default 0)\n\n**Approach Miss Patterns**
(Distribution):\n- `approach_miss_left_pct`, `approach_miss_right_pct`,
`approach_miss_short_pct`, `approach_miss_long_pct` (NUMERIC(5,2))\n\n**Trending (Added in
Migration 053)**:\n- `last_5_average`, `last_10_average` (NUMERIC(5,2))\n- `improvement_trend`
(NUMERIC(5,2)) - Positive = improving\n- `trend_direction` (TEXT: 'improving', 'stable',
'declining')\n- `sg_total_per_round`, `sg_tee_per_round`, `sg_approach_per_round`,
`sg_around_green_per_round`, `sg_putting_per_round` (NUMERIC(5,2))\n- `rounds_this_season`
(INTEGER, default 0)\n- `season_start_date` (DATE)\n- `is_stale` (BOOLEAN, default FALSE)\n-
`next_refresh_due` (TIMESTAMPTZ)\n- `round_ids_included` (UUID[]) - Array of round IDs in
calculation\n\n**Metadata**:\n- `last_round_date` (DATE)\n- `rounds_in_calculation` (INTEGER,
default 0)\n- `calculation_period_start`, `calculation_period_end` (DATE)\n- `engine_version`
(TEXT, default 'v2')\n- `created_at`, `updated_at` (TIMESTAMPTZ)\n\n**Indexes**:\n-
`idx_golf_player_stats_cache_player` (player_id)\n- `idx_golf_player_stats_cache_updated`
(updated_at DESC)\n- `idx_golf_player_stats_cache_stale` (is_stale) WHERE is_stale = TRUE\n-
`idx_golf_player_stats_cache_refresh` (next_refresh_due) WHERE next_refresh_due IS NOT

```


NULL\n\n**RLS**:\n- Players can view own stats\n- Coaches can view team player stats\n- Coaches can manage team stats cache\n\n---\n\n### 2.2 golf_round_stats_cache\n\n**Location**:
Migration 040_golf_shot_system.sql\n\n**Purpose**:
Per-round calculated stats for quick round review without recalculation\n\n**Primary Key**:
UUID\n\n**Unique Constraint**:
round_id\n\n**FK**:
round_id → golf_rounds(id) ON DELETE CASCADE, player_id → golf_players(id) ON DELETE CASCADE\n\n**Basic Scoring**:
\n- `total\_score` (INTEGER)\n- `score\_to\_par` (INTEGER)\n- `front\_nine`, `back\_nine` (INTEGER)\n\n**Strokes Gained for this Round**:
\n- `strokes\_gained\_total`, `strokes\_gained\_tee`, `strokes\_gained\_approach`, `strokes\_gained\_around\_green`, `strokes\_gained\_putting` (NUMERIC(5,2))\n\n**Driving this Round**:
\n- `fairways\_hit`, `fairways\_total` (INTEGER)\n- `driving\_distance\_avg` (NUMERIC(6,1))\n\n**GIR this Round**:
\n- `greens\_hit`, `greens\_total` (INTEGER)\n\n**Putting this Round**:
\n- `total\_putts` (INTEGER)\n- `one\_putts`, `three\_putts` (INTEGER)\n\n**Scrambling this Round**:
\n- `scrambles\_converted`, `scramble\_attempts` (INTEGER)\n- `sand\_saves`, `sand\_attempts` (INTEGER)\n\n**Scoring Distribution**:
\n- `eagles`, `birdies`, `pars`, `bogeys`, `double\_bogeys`, `triple\_plus` (INTEGER, defaults 0)\n\n**Penalties**:
\n- `penalty\_strokes` (INTEGER, default 0)\n\n**Full Stats Storage**:
\n- `detailed\_stats` (JSONB) - Complete GolfStats output\n\n**Timestamps**:
\n- `created\_at`, `updated\_at` (TIMESTAMPZ)\n\n**Indexes**:
\n- `idx\_golf\_round\_stats\_cache\_round` (round_id)\n- `idx\_golf\_round\_stats\_cache\_player` (player_id)\n- `idx\_golf\_round\_stats\_cache\_player\_date` (player_id, created_at DESC)\n\n**RLS**:
\n- Players can view own round stats\n- Coaches can view team round stats\n- Coaches can manage round stats cache\n\n---\n\n### 2.3 golf_putting_tendencies\n\n**Location**:
Migration 040_golf_shot_system.sql\n\n**Purpose**:
Putting pattern analysis by break type\n\n**Primary Key**:
UUID\n\n**Unique Constraint**:
player_id\n\n**FK**:
player_id → golf_players(id) ON DELETE CASCADE\n\n**Overall**:
\n- `total\_putts\_analyzed` (INTEGER, default 0)\n- `make\_percentage\_overall` (NUMERIC(5,2))\n\n**By Break Type**:
\n- `left\_to\_right\_attempts`, `left\_to\_right\_made` (INTEGER, default 0)\n- `left\_to\_right\_pct` (NUMERIC(5,2))\n- `right\_to\_left\_attempts`, `right\_to\_left\_made` (INTEGER, default 0)\n- `right\_to\_left\_pct` (NUMERIC(5,2))\n- `straight\_attempts`, `straight\_made` (INTEGER, default 0)\n- `straight\_pct` (NUMERIC(5,2))\n\n**Miss Tendencies**:
\n- `miss\_left\_percentage`, `miss\_right\_percentage`, `miss\_short\_percentage`, `miss\_long\_percentage` (NUMERIC(5,2))\n\n**Distance Tendencies**:
\n- `avg\_distance\_on\_makes\_ft`, `avg\_distance\_on\_misses\_ft` (NUMERIC(5,1))\n\n**Pressure Putting (Inside 5 feet)**:
\n- `short\_putt\_attempts`, `short\_putt\_made` (INTEGER, default 0)\n- `short\_putt\_pct` (NUMERIC(5,2))\n\n**Metadata**:
\n- `last\_calculated` (TIMESTAMPZ)\n- `rounds\_in\_sample` (INTEGER, default 0)\n- `created\_at`, `updated\_at` (TIMESTAMPZ)\n\n**Indexes**:
\n- `idx\_golf\_putting\_tendencies\_player` (player_id)\n\n**RLS**:
\n- Players can view own tendencies\n- Coaches can view team tendencies\n- Coaches can manage tendencies\n\n---\n\n### 3. DETAILED SHOT TRACKING TABLES\n\n### 3.1 putt_details\n\n**Location**:
Migration 064_putt_approach_details.sql\n\n**Purpose**:
Detailed putt miss classification and break analysis\n\n**Primary Key**:
UUID\n\n**FK**:
shot_id → golf_shots(id) ON DELETE CASCADE (UNIQUE)\n\n**Columns**:
\n- `shot\_id` (UUID, FK, UNIQUE) ★\n- `miss\_tags` (TEXT[], default '{}') - Array: 'low', 'high', 'short'\n- `break\_direction` (TEXT: 'left_to_right', 'right_to_left', 'straight', 'multiple')\n- `estimated\_break\_inches` (INTEGER, 0-48)\n- `distance\_feet` (NUMERIC, 0-200)\n- `made` (BOOLEAN, NOT NULL, default FALSE)\n- `created\_at`, `updated\_at` (TIMESTAMPZ)\n\n**Indexes**:
\n- `idx\_putt\_details\_shot\_id` (shot_id)\n\n**RLS**:
\n- Players can view/edit/delete own putts\n- Coaches can view team putts\n\n---\n\n### 3.2 approach_miss_details\n\n**Location**:
Migration 064_putt_approach_details.sql\n\n**Purpose**:
Detailed approach miss direction and lie tracking\n\n**Primary Key**:
UUID\n\n**FK**:
shot_id → golf_shots(id) ON DELETE CASCADE (UNIQUE)\n\n**Columns**:
\n- `shot\_id` (UUID, FK, UNIQUE) ★\n- `miss\_direction` (TEXT: 'short', 'long', 'left', 'right', 'short_left', 'short_right', 'long_left', 'long_right')\n- `lie\_type` (TEXT: 'fairway', 'rough', 'sand', 'recovery', 'hazard')\n- `distance\_from\_green\_yards` (NUMERIC, >= 0)\n- `created\_at`, `updated\_at` (TIMESTAMPZ)\n\n**Indexes**:
\n- `idx\_approach\_miss\_details\_shot\_id` (shot_id)\n\n**RLS**:
\n- Players can view/edit/delete own approach misses\n- Coaches can view team approach misses\n\n---\n\n### 4. ANALYTICS & INSIGHTS TABLES\n\n### 4.1 golf_round_reviews\n\n**Location**:
Migration 039_golf_analytics_tables.sql\n\n**Purpose**:
CoachHelm AI round analysis and review\n\n**Primary Key**:
UUID\n\n**Unique Constraint**:
CoachHelm AI round analysis and review

```

round_id\n\n**Round Summary**:\n- `round_id` (UUID, FK) ★\n- `player_id` (UUID, FK) ★\n-
`round_score`, `round_score_to_par` (INTEGER)\n- `scoring_avg_before`, `scoring_avg_after`
(NUMERIC(4,2))\n\n**AI Analysis**:\n- `highlights` (JSONB, default '[]')\n- `areas_to_review`
(JSONB, default '[]')\n- `round_stats` (JSONB)\n- `patterns_detected` (JSONB, default
'[]')\n\n**Text Summaries**:\n- `summary`, `primary_takeaway`, `next_practice_priority`
(TEXT)\n\n**Coach Interaction**:\n- `coach_notes` (TEXT)\n- `coach_viewed_at` (TIMESTAMPTZ)\n-
`shared_with_coach` (BOOLEAN, default FALSE)\n- `shared_at` (TIMESTAMPTZ)\n\n**Metadata**:\n-
`engine_version` (TEXT, default 'v2')\n- `created_at`, `updated_at`
(TIMESTAMPTZ)\n\n**Indexes**:\n- `idx_golf_round_reviews_player` (player_id)\n-
`idx_golf_round_reviews_round` (round_id)\n\n**RLS**:\n- Players can view own reviews\n-
Coaches can view shared team reviews\n- Coaches can manage team reviews\n\n---\n\n### 4.2
golf_coach_insights\n\n**Location**:\nMigration 039_golf_analytics_tables.sql\n\n**Purpose**:\nCoachHelm proactive alerts for coaches\n\n**Primary Key**:\nUUID\n\n**FKs**:\ncoach_id →
golf_coaches(id) ON DELETE CASCADE, player_id → golf_players(id) ON DELETE CASCADE, team_id →
golf_teams(id) ON DELETE CASCADE\n\n**Classification**:\n- `insight_type` (TEXT, CHECK):
'performance_decline', 'performance_improvement', 'pattern_detected',
'practice_recommendation', 'roster_alert', 'qualifying_watch', 'attendance_concern',
'milestone_reached', 'comparison_insight'\n- `title`, `content` (TEXT)\n- `priority` (TEXT,
CHECK: 'low', 'medium', 'high', 'urgent', default 'medium')\n\n**Status Tracking**:\n-
`status` (TEXT, CHECK: 'active', 'acknowledged', 'dismissed', 'resolved', default 'active')\n-
`acknowledged_at`, `dismissed_at`, `resolved_at` (TIMESTAMPTZ)\n- `dismissed` (BOOLEAN,
default FALSE)\n\n**Additional Data**:\n- `metadata` (JSONB)\n- `created_at`, `updated_at`
(TIMESTAMPTZ)\n\n**Indexes**:\n- `idx_golf_coach_insights_coach` (coach_id)\n-
`idx_golf_coach_insights_player` (player_id)\n- `idx_golf_coach_insights_status` (status)
WHERE status = 'active'\n\n**RLS**:\n- Coaches can view their own insights\n- Coaches can
manage their insights\n\n---\n\n### 4.3 golf_insight_generation_log\n\n**Location**:\nMigration
039_golf_analytics_tables.sql\n\n**Purpose**:\nAudit log for insight generation runs\n\n**Primary
Key**:\nUUID\n\n**FKs**:\nteam_id → golf_teams(id) ON DELETE CASCADE, player_id →
golf_players(id) ON DELETE CASCADE\n\n**Columns**:\n- `insight_type` (TEXT)\n-
`rounds_analyzed`, `insights_generated` (INTEGER)\n- `engine_version` (TEXT)\n- `duration_ms`
(INTEGER)\n- `created_at` (TIMESTAMPTZ)\n\n**Indexes**:\n- `idx_golf_insight_log_team`
(team_id)\n- `idx_golf_insight_log_created` (created_at DESC)\n\n**RLS**:\nCoaches can view
team logs\n\n---\n\n### 5. ADDITIONAL STATS TRACKING TABLES\n\n### 5.1
golf_attendance_summary\n\n**Location**:\nMigration 039_golf_analytics_tables.sql\n\n**Purpose**:\nAggregated attendance tracking\n\n**Primary Key**:\nUUID\n\n**Unique Constraint**:\n(player_id,
team_id, period_start_date, period_end_date)\n\n**FKs**:\nplayer_id → golf_players(id), team_id
→ golf_teams(id)\n\n**Columns**:\n- `total_events`, `attended_count`, `absent_count`,
`excused_count` (INTEGER, defaults 0)\n- `attendance_percentage` (NUMERIC(5,2))\n-
`period_start_date`, `period_end_date` (DATE)\n- `created_at`, `updated_at`
(TIMESTAMPTZ)\n\n**Indexes**:\n- `idx_golf_attendance_summary_player` (player_id)\n-
`idx_golf_attendance_summary_team` (team_id)\n\n**RLS**:\nPlayers can view own, coaches can
view team\n\n---\n\n### 5.2 golf_academic_exclusions\n\n**Location**:\nMigration
039_golf_analytics_tables.sql\n\n**Purpose**:\nAcademic ineligibility tracking\n\n**Primary Key**:\nUUID\n\n**FKs**:\nplayer_id → golf_players(id), excluded_by →
golf_coaches(id)\n\n**Columns**:\n- `start_date`, `end_date` (DATE)\n- `reason` (TEXT)\n-
`created_at`, `updated_at` (TIMESTAMPTZ)\n\n**Indexes**:\n-
`idx_golf_academic_exclusions_player` (player_id)\n- `idx_golf_academic_exclusions_dates`
(start_date, end_date)\n\n---\n\n### 5.3 golf_event_exclusions\n\n**Location**:\nMigration
039_golf_analytics_tables.sql\n\n**Purpose**:\nPlayer exclusion from specific events\n\n**Primary
Key**:\nUUID\n\n**Unique Constraint**:\n(player_id, event_id)\n\n**FKs**:\nplayer_id →
golf_players(id), event_id → golf_events(id), excluded_by →
golf_coaches(id)\n\n**Columns**:\n- `reason` (TEXT)\n- `created_at`
(TIMESTAMPTZ)\n\n**Indexes**:\n- `idx_golf_event_exclusions_event` (event_id)\n-
`idx_golf_event_exclusions_player` (player_id)\n\n---\n\n### 6. COURSES & HOLES\n\n### 6.1
golf_courses\n\n**Location**:\nMigration 022_golf_courses.sql\n\n**Primary Key**:\nUUID\n\n**Unique
Constraint**:\n(name, city, state, created_by)\n\n**Columns**:\n- `name`, `city`, `state`,
`country` (TEXT, country default 'USA')\n- `course_rating` (DECIMAL(4,1))\n- `slope_rating`
(INTEGER)\n- `default_tee_name`, `default_tee_color` (TEXT)\n- `total_yardage`, `total_par`

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(INTEGER)\n- `created_by` (UUID, FK → auth.users)\n- `is_public` (BOOLEAN, default FALSE)\n- `created_at`, `updated_at` (TIMESTAMPZ)\n\n**Indexes**:\n- `idx_golf_courses_created_by` (created_by)\n- `idx_golf_courses_name` (name)\n\n---\n\n6.2  
golf_course_holes\n\n**Location**: Migration 022_golf_courses.sql\n\n**Primary Key**:  
UUID\n\n**Unique Constraint**: (course_id, hole_number)\n\n**FKs**: course_id → golf_courses(id)  
ON DELETE CASCADE\n\n**Columns**:\n- `hole_number` (INTEGER, 1-18)\n- `par` (INTEGER, 3-6)\n- `yardage` (INTEGER, > 0)\n- `handicap_index` (INTEGER, 1-18, optional)\n\n**Indexes**:\n- `idx_golf_course_holes_course_id` (course_id)\n- `idx_golf_course_holes_course` (course_id)\n\n---\n\n6.3 golf_course_tees\n\n**Location**: Migration 022_golf_courses.sql\n\n**Primary Key**: UUID\n\n**Unique Constraint**: (course_id, tee_name)\n\n**FKs**: course_id → golf_courses(id) ON DELETE CASCADE\n\n**Columns**:\n- `tee_name` (TEXT)\n- `tee_color` (TEXT)\n- `course_rating` (DECIMAL(4,1))\n- `slope_rating` (INTEGER)\n- `total_yardage` (INTEGER)\n- `hole_yardages` (JSONB) - Per-hole yardages for this tee\n\n---\n\n7. GOLF ENUMS (Migration 001_extensions_and_enums.sql)\n\n**Golf-Specific Enums**:\n- `golf_player_year`: 'freshman', 'sophomore', 'junior', 'senior', 'fifth_year', 'graduate'\n- `golf_player_status`: 'active', 'injured', 'redshirt', 'inactive'\n- `golf_round_type`: 'tournament', 'qualifier', 'practice', 'casual'\n- `golf_round_status`: 'in_progress', 'paused', 'completed', 'abandoned'\n- `golf_event_type`: 'practice', 'tournament', 'qualifier', 'meeting', 'travel', 'other'\n- `golf_qualifier_status`: 'upcoming', 'in_progress', 'completed'\n- `golf_alert_sensitivity`: 'aggressive', 'balanced', 'conservative'\n- `golf_shot_club`: 'driver', '3_wood', '5_wood', '7_wood', 'hybrid', '2_iron'-'9_iron', 'pw', 'gw', 'sw', 'lw', 'putter'\n- `golf_shot_result`: 'excellent', 'good', 'average', 'poor', 'miss'\n- `golf_miss_direction`: 'left', 'right', 'short', 'long', 'left_short', 'left_long', 'right_short', 'right_long'\n\n---\n\n8. SEED DATA\n\n**No dedicated seed file found** (`supabase/seed.sql` does not exist for golf).\n\nThe codebase relies on:\n1. User-generated data through the application UI\n2. Database migrations creating empty tables\n3. Optional manual test data during development\n\nTypeScript types in `/src/lib/types/golf.ts` support creating test data programmatically.\n\n---\n\n9. RLS (Row Level Security) POLICIES\n\nComprehensive RLS Matrix:\n\n| Table | Players (Own) | Players (Others) | Coaches | Notes |\n|-----|-----|-----|-----|-----|\n| View team | Basic UNIQUE constraint on user_id | View own | View team | Via player_id FK | Nested: via round-player |\n| View golf_holes | View own | View team | Nested: via hole-round-player |\n| View golf_shots | View own | View team | Nested: via hole-round-player |\n| View golf_player_stats_cache | View own | View team | Player/Coach policies |\n| View golf_round_stats_cache | View own | View team | Player/Coach policies |\n| View golf_putting_tendencies | View own | View team | Player/Coach policies | putt_details | View own | View team | Nested RLS to shot-round-player | approach_miss_details | View own | View team | Nested RLS to shot-round-player | View own | View shared | Shared_with_coach flag required | View own | Coach-only access | Manage team | Organization-based coach access | Most golf stats tables have *potentially dangerous* nested RLS policies (through multiple FK joins). Migrations 20260119-20260204 fixed infinite recursion issues multiple times.\n\n10. PERFORMANCE INDEXES (Migration 050)\n\nHigh-Impact Composite Indexes for stats queries:\n1. Round Retrieval (player dashboard):\n- idx_golf_rounds_player_status_date(player_id, status, round_date DESC)\n- idx_golf_rounds_in_progress(player_id, updated_at DESC)\n2. Shot Data Aggregation (stats calculator):\n- idx_golf_shots_round_hole_shot(round_id, hole_number, shot_number)\n- idx_golf_shots_round_type(round_id, shot_type)\n3. Calendar/Events:\n- idx_golf_events_team_start_time(team_id, start_time)\n4. Team Members (very frequent):\n- idx_golf_team_members_player(player_id, team_id) - Used on ~every request\n5. Qualifier Leaderboards:\n- idx_golf_qualifier_entries_leaderboard(qualifier_id, score NULLS LAST)\n\n11. CALCULATED STATS GENERATION\n\nServer-Side Calculation (referenced but not fully visible in migrations):\n- calculate_round_stats() - Aggregate hole scores into round summary\n- update_player_stats_cache_enhanced() - Trigger for trend calculations\n- mark_player_stats_stale() - Invalidate cache on data changes\n- get_player_stats_summary()
```

- Fetch cache with freshness check\n\n**Stats are calculated by**:\n- Golf stats calculator engine: `/src/lib/utils/golf-stats-calculator-shots.ts` (1693 lines)\n- Supports 100+ metrics including strokes gained, putting analysis, approach efficiency\n- Uses `golf\_shots` table with distance/lie details for precision\n\n---\n\n## 12. KEY OBSERVATIONS\n\n### Strengths:\n\n✅ Comprehensive shot-level tracking (golf_shots table)\n✅ Multiple cache layers for performance (player_stats_cache, round_stats_cache)\n✅ Detailed putt & approach miss tracking (putt_details, approach_miss_details)\n✅ Strategic composite indexes for common query patterns\n✅ Separation of concerns (cache vs. raw data)\n✅ Strong RLS on stats cache tables\n\n### Challenges:\n\n⚠️ Nested RLS policies with potential for infinite recursion (see migrations 20260119+)\n⚠️ No formal seed data for testing\n⚠️ Complex trigger chains for cache invalidation\n⚠️ Coach access determined via organization_id on golf_teams (not always present)\n⚠️ golf_shots table has many optional columns (data quality issues possible)\n\n### Data Quality Depends On:\n\n- Proper `distance\_to\_hole\_before`, `distance\_to\_hole\_after`, `club\_type` population in golf_shots\n- Regular cache refresh via trigger functions\n- Correct round_status ('completed' vs 'in_progress') classification\n\nThis schema supports advanced golf analytics including strokes gained, putting patterns, approach efficiency, and AI-driven coaching insights through

```
CoachHelm."}], "stop_reason": null, "stop_sequence": null, "usage": {"input_tokens": 0, "cache_creation_input_tokens": 3805, "cache_read_input_tokens": 142074, "cache_creation": {"ephemeral_5m_input_tokens": 0, "ephemeral_1h_input_tokens": 3805}, "output_tokens": 1, "service_tier": "standard", "inference_geo": "not_available"}}, "requestId": "req_011CXqYG98iNwWLaGfchsuvW", "type": "assistant", "uuid": "0a9754e8-6085-4473-b68d-cd188ef99998", "timestamp": "2026-02-05T20:19:18.757Z"}
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